BASIC PRINCIPLES OF MEDICINE 1: FOUNDATIONS TO MEDICINE

LECTURE FTM 45
Karyotypes and Aneuploidy

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Pre-Lecture - Required Reading and Objectives

- **Robbins Essential Pathology,** Chapter 6, section on Cytogenetic Disorders involving Autosomes and Sex Chromosomes:
  - Genetic Diseases - Robbins Essential Pathology - ClinicalKey Student

SOM.MKIII.BPM1.1.FTM.3.GNET.0121	Analyze the abnormal human karyotype and define the terms euploidy and aneuploidy. Use disorders and syndromes such as Down, Patau, Edward, Turner, Klinefelter as examples

Recommended Reading and Objectives

- **Human Genetics and Genomics, Korf and Irons, 4th Edition Chapter 6**
  - **You can find the following pages and chapters on your Sakia site under:**
  - Resources/General Information/Book Access

SOM.MKIII.BPM1.1.FTM.3.GNET.0120	Analyze the normal human karyotype and how it is obtained

Objectives

SOM.MKIII.BPM1.1.FTM.3.GNET.0120	Analyze the normal human karyotype and how it is obtained
SOM.MKIII.BPM1.1.FTM.3.GNET.0121	Analyze the abnormal human karyotype and define the terms euploidy and aneuploidy. Use disorders and syndromes such as Down, Patau, Edward, Turner, Klinefelter as examples
SOM.MKIII.BPM1.1.FTM.3.GNET.0122	Analyze meiosis to explain how chromosomal aberrations are formed
SOM.MKIII.BPM1.1.FTM.3.GNET.0123	Discuss mechanisms that might lead to mosiacism for chromosomal aneuploid conditions

Methods of Chromosome Analysis

Laboratory techniques used for karyotyping

Table 3-1. Development of Methodologies for Cytogenetics

Decade	Development	Examples of Application
1950-1960s	Reliable methods for chromosome preparations	Chromosome number determined to be 46 (1956) and Philadelphia chromosome identified as t(9;22) (1960)
1970s	Giemsa chromosome banding	Mapping of *RB1* gene to chromosome 13q14 by identification of deleted chromosomal region in patients with retinoblastoma (1976)
1980s	Fluorescent in-situ hybridization (FISH)	Interphase FISH for rapid detection of Down syndrome (1994) Spectral karyotyping for whole genome chromosome analysis (1996)
1990s	Comparative genomic hybridization (CGH)	Mapping genomic imbalances in solid tumors (1992)
2000s	Array CGH	Analysis of constitutional rearrangements; e.g., identification of ~5 Mb deletion in a patient with CHARGE syndrome that led to identification of the gene (2004)

CHARGE, **c**oloboma of the eye, **h**eart defects, **a**tresia of the choanae, **r**etardation of growth and/or development, **g**enital and/or urinary abnormalities, and **e**ar abnormalities and deafness.

Emery Table 3.1

Chromosomal abnormalities detectable by cytogenetics

- Chromosomal abnormalities account for a large number of genetic malformation syndromes, spontaneous abortions, and infertility cases

- **Numerical Chromosomal abnormalities may be a result of:**
    - Non-disjunction during meiosis
    - Non-disjunction during mitosis
    - Mosaicism

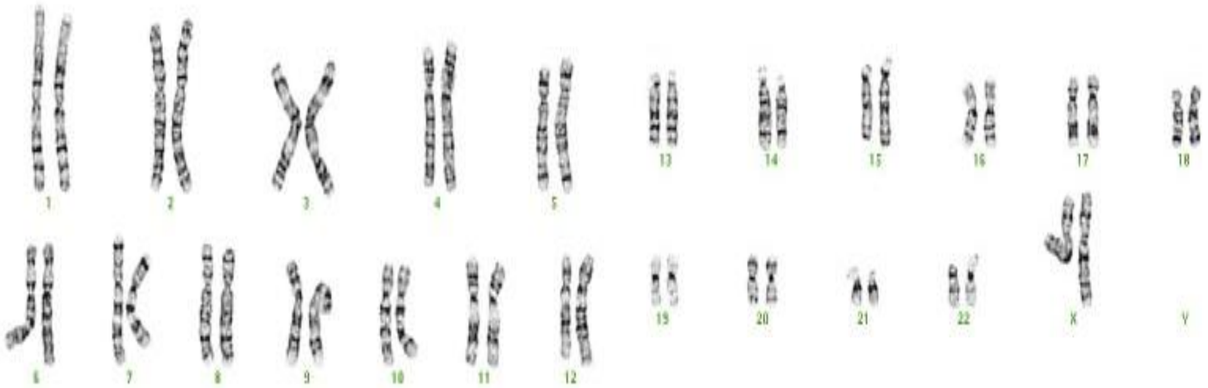
This lecture

- **Structural Chromosomal abnormalities may arise from:**
    - Deletions
    - Duplication
    - Inversions
    - Translocations
    - Isochromosomes
    - Ring Chromosomes

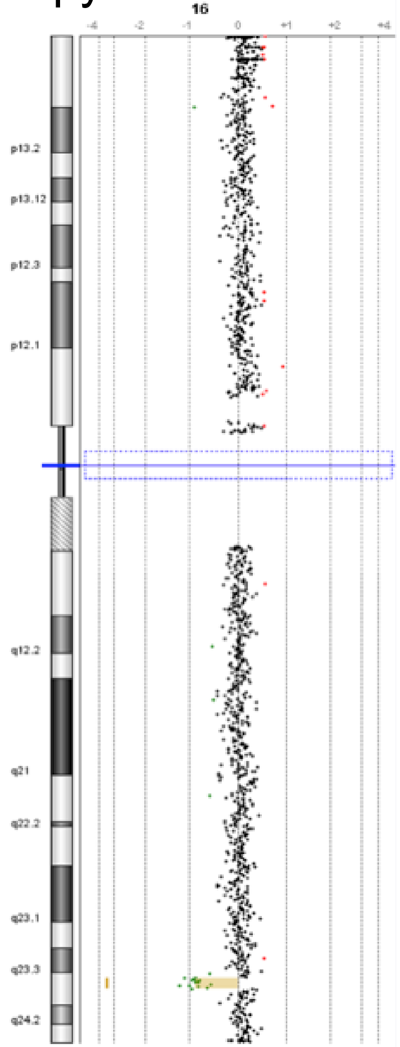
Next lecture

Cytogenetic Testing

Karyotype, giemsa staining

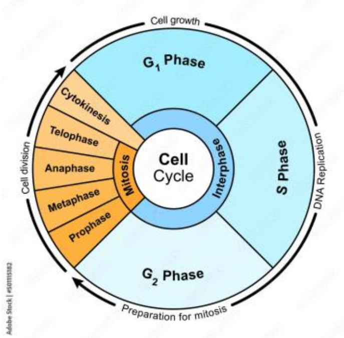


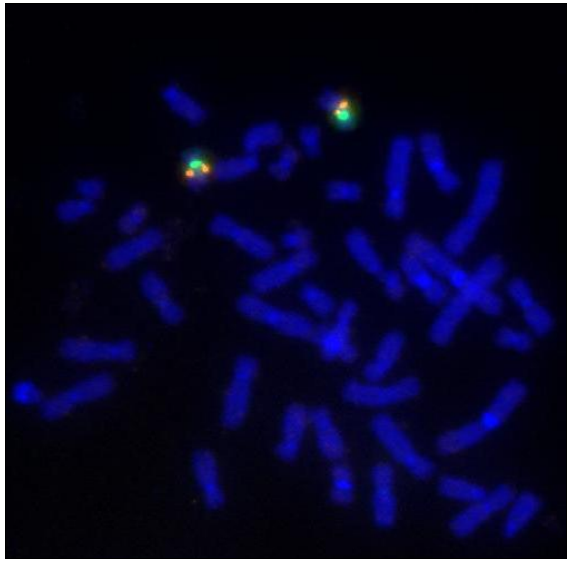
CGH Microarray
Copy Number Variants

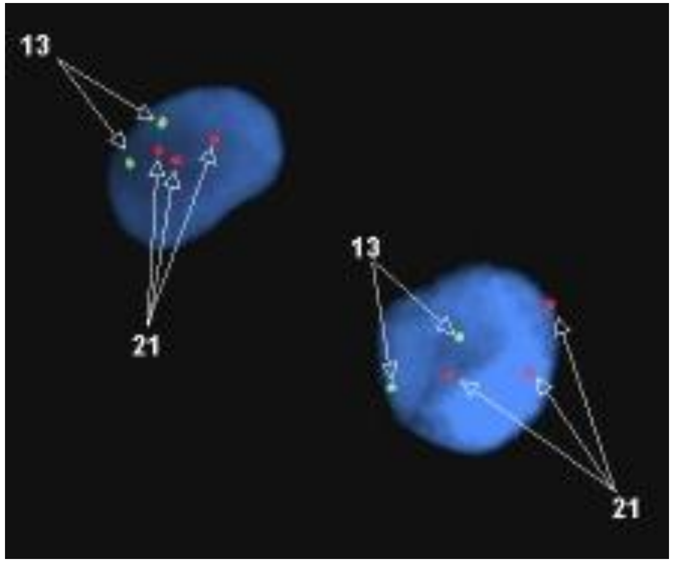


FISH fluorescent probes

Interphase FISH fluorescent probes







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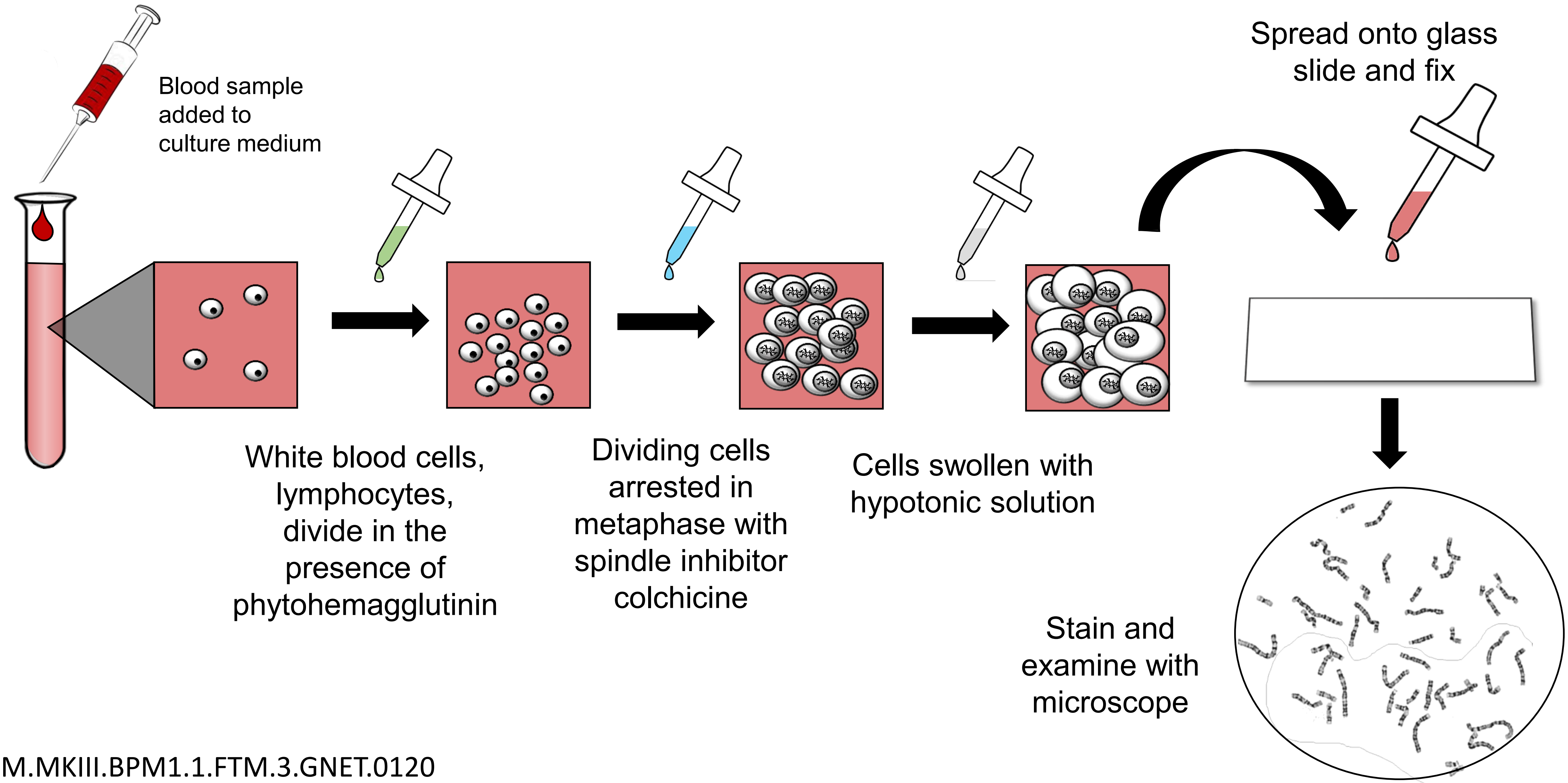
Cytogenetics: Generating a karyotype



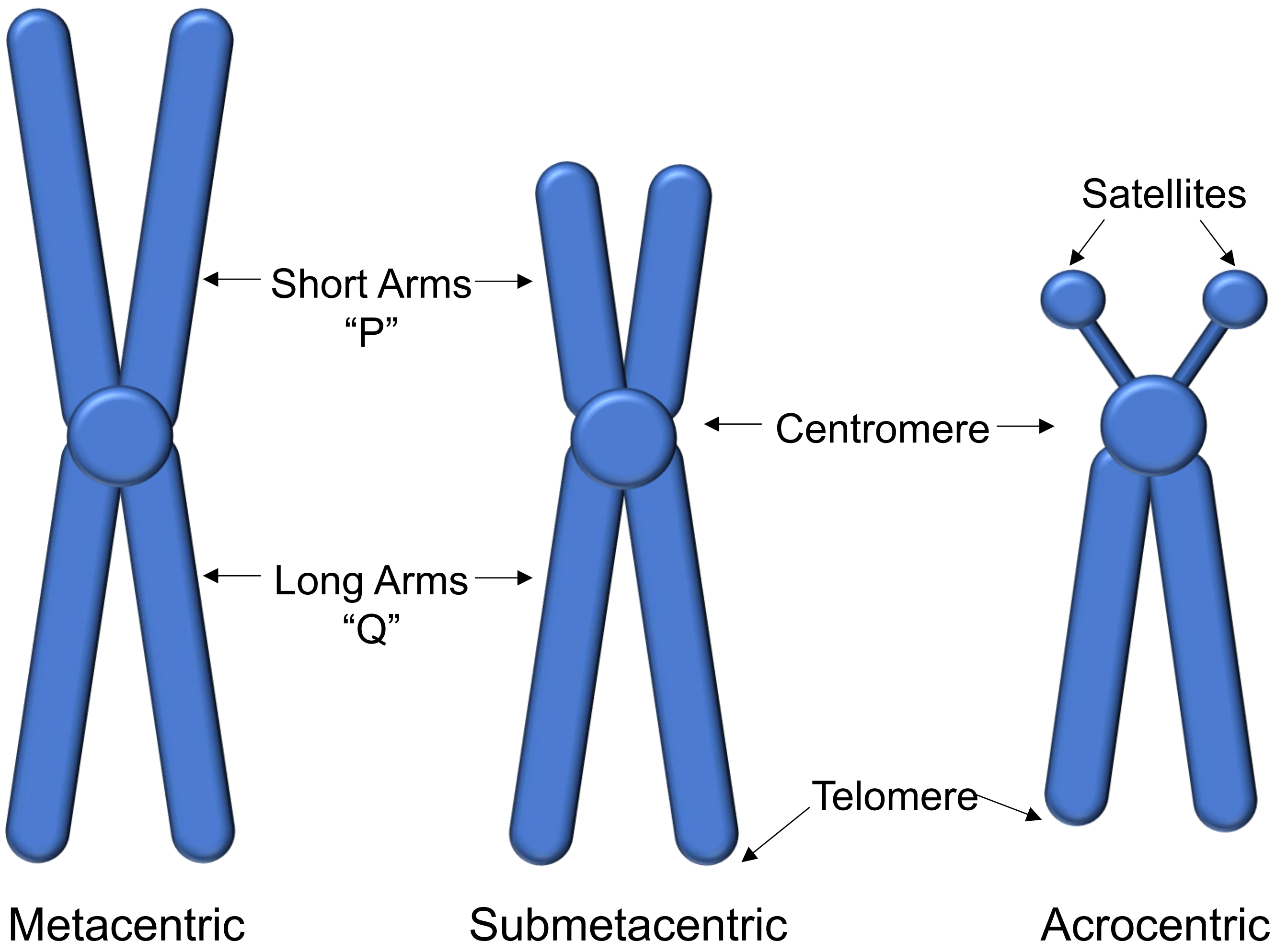
Karyotype

- Karyotype is analyzed by microscopy
  - Only relatively large alterations will be observable by karyotype
  - Banding pattern must be altered

- Can detect deletions, duplications, inversions, translocations, other abnormalities

- Deletions larger than 5 Mb are usually visible by standard karyotype (this is by microscopy)
  - Deletions that are not visible by karyotype are called “microdeletions”

Classifying Chromosomes by Structure



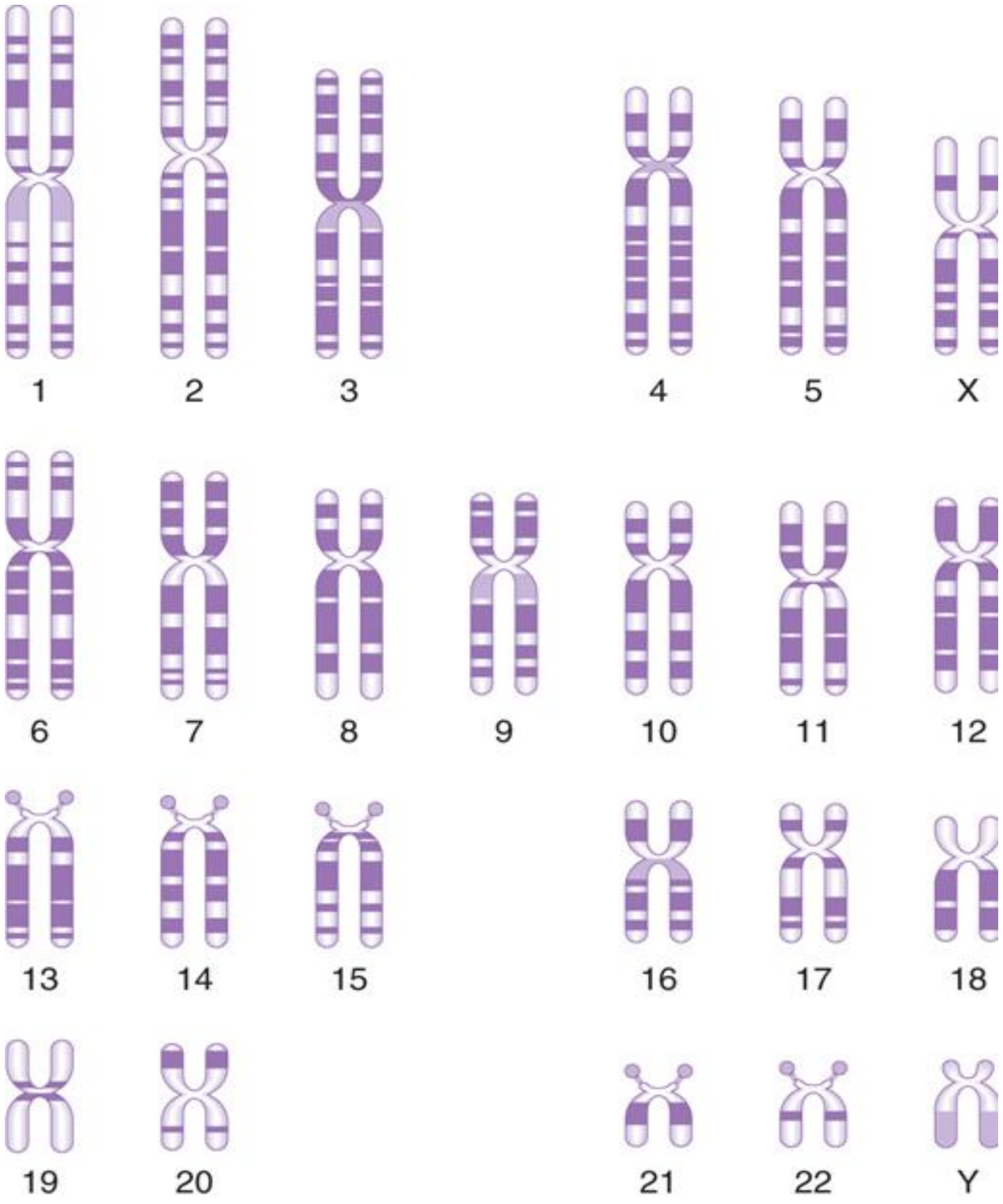
1) Total length of the chromosome

2) Position of the centromere

3) Presence of satellites

Classifying Chromosomes by G-banding pattern

Sister chromatids,
Only one chromosome each



Ideogram of the G-banding pattern of metaphase chromosomes stained with Giemsa

Resolution: can't detect deletions or insertions smaller than 5Mb

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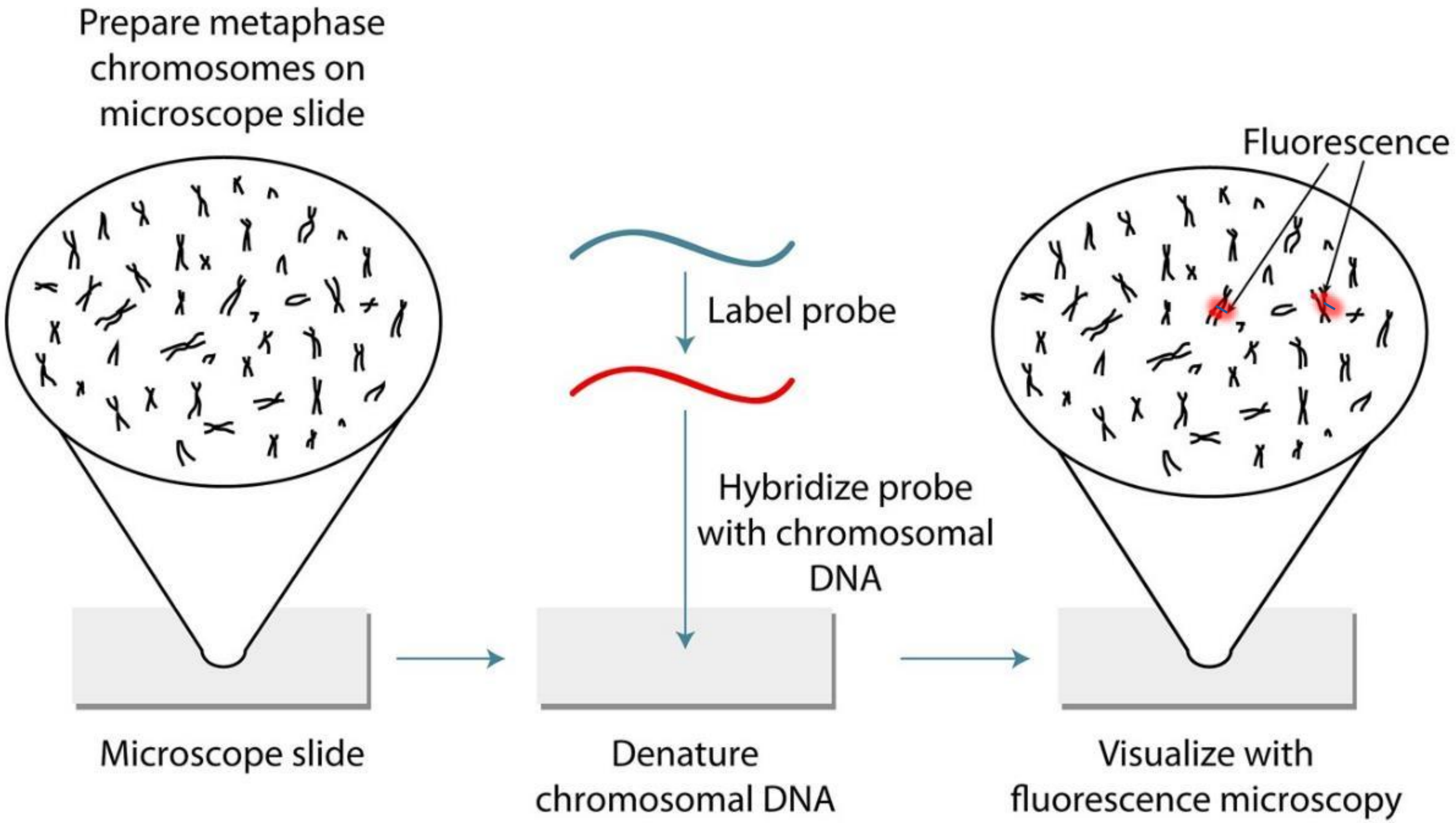
Fluorescence *in situ* Hybridization (FISH)

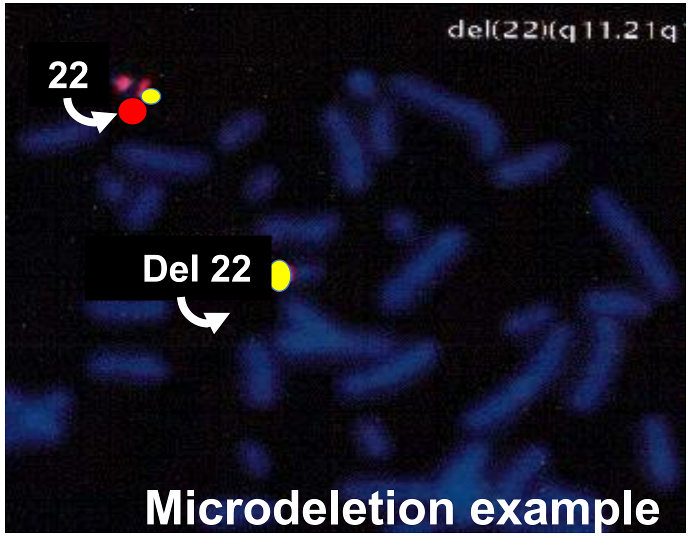


FISH

- **FISH probes are big;** *ASO probes are small*
  - Typically to get a good signal, they are about **100 kb** or bigger
  - This allows a lot of fluorescent tags to be packed onto the probe

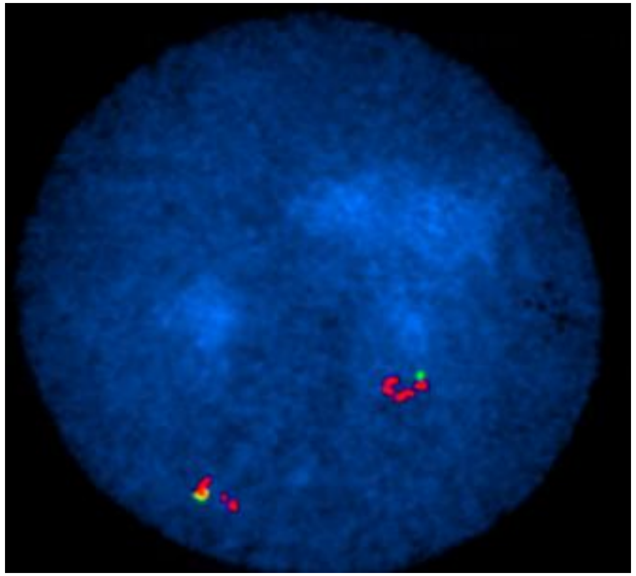
- With the correct probe FISH can detect:
  - Numerical chromosome aberrations
  - Deletions
  - Translocations
  - Gene amplification (very important in cancer characterization)

- FISH may detect many "microdeletions" because FISH is of higher resolution than G-band karyotype
  - As long as the deletion is larger than the probe being used, the deletion can be detected

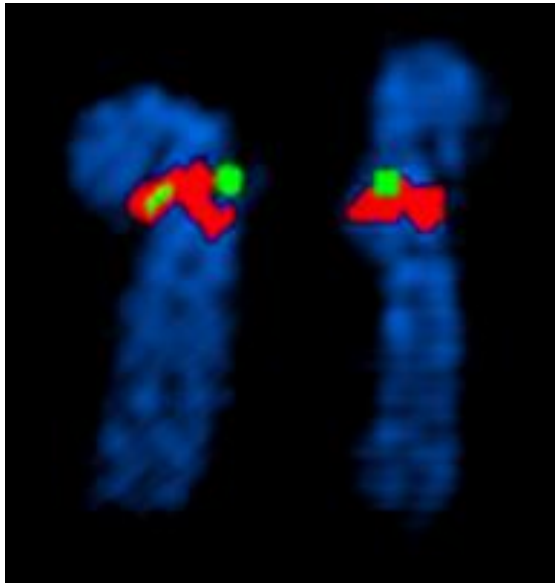


FISH

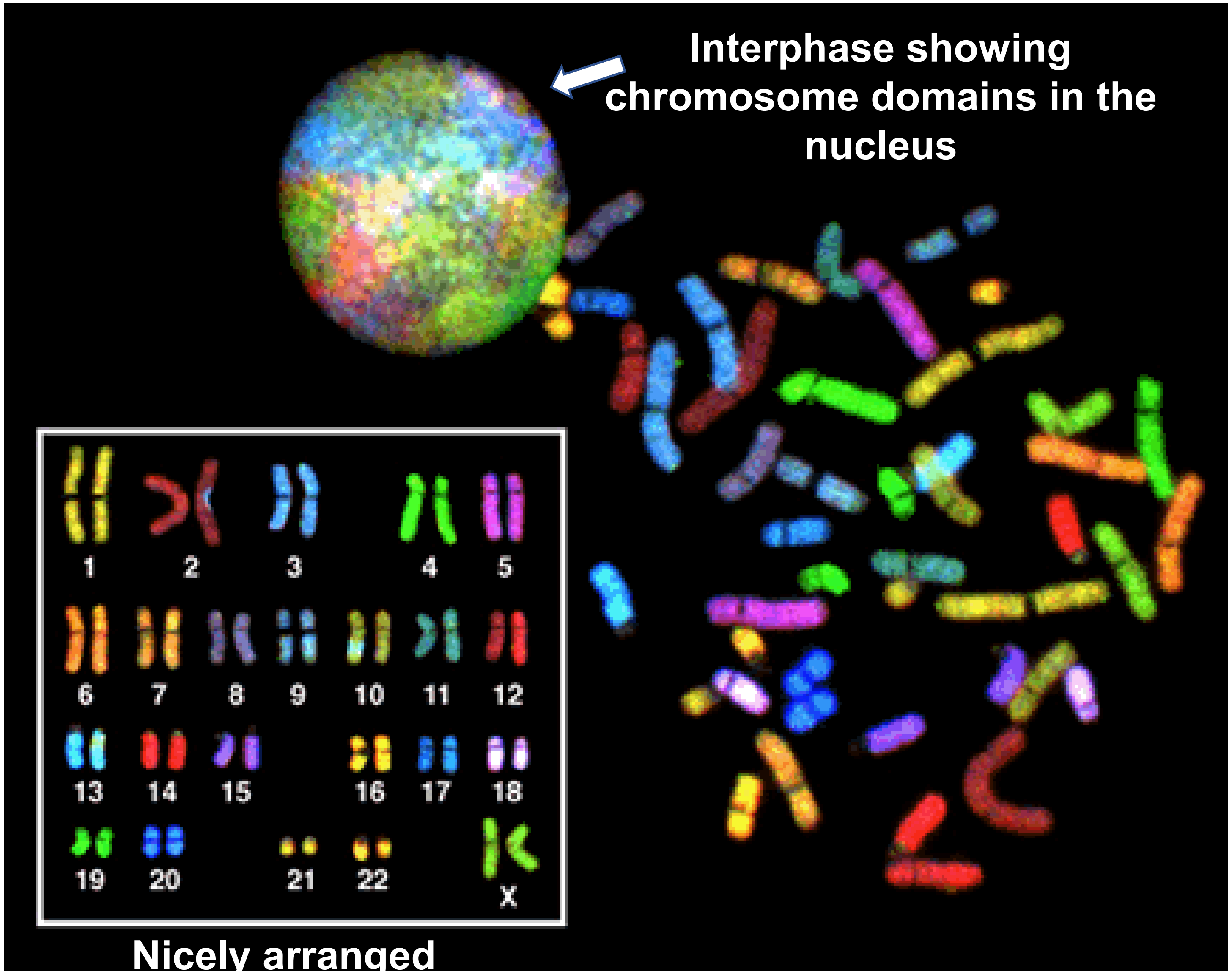
- Interphase stage FISH
  - Rapid test
  - Avoids need for cell culture
  - Chromosomes are decondensed in the nucleus



- Metaphase stage FISH
  - Must first culture the cells
  - Usually done after a G-band karyotype (condensed metaphase chromsomes).
  - Gives location information because the banding pattern would be known



Chromosome painting



- Chromosome painting, spectral karyotype, SKY FISH
  - Each chromosome probed so that it fluoresces a different color
- Can show translocations and rearrangements
- Very useful in **cancer genetics**
  - Genomic instability in advanced cancers

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Chromosomal abnormalities

- Chromosomal abnormalities account for a large number of spontaneous abortions
  - (noninduced embryonic or fetal death or passage of products of conception before 20 weeks gestation)
- Chromosomal abnormalities may be
  - Numerical
  - Structural

Numerical Chromosome Abnormalities

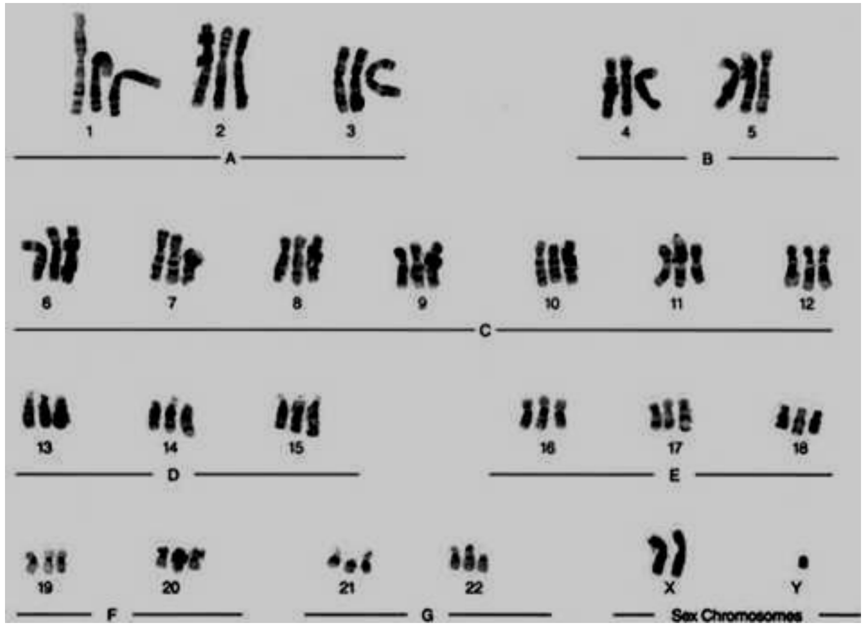
Euploidy (polyploidy) (Number of chromosomes is in multiples of 23)

- Diploidy: Normal human compliment of chromosomes, two of each (46)
- Triploidy: contain **three** copies of each chromosome (69)
  - Not compatible with life
- Tetraploidy: Contain **four** copies of each chromosome (92)
  - Lethal

Aneuploidy

- Monosomy: Loss of a chromosome
  - Not compatible with life (Except monosomy X: Turner syndrome)
- Trisomy: Presence of an additional chromosome (only 3 are compatible with life)

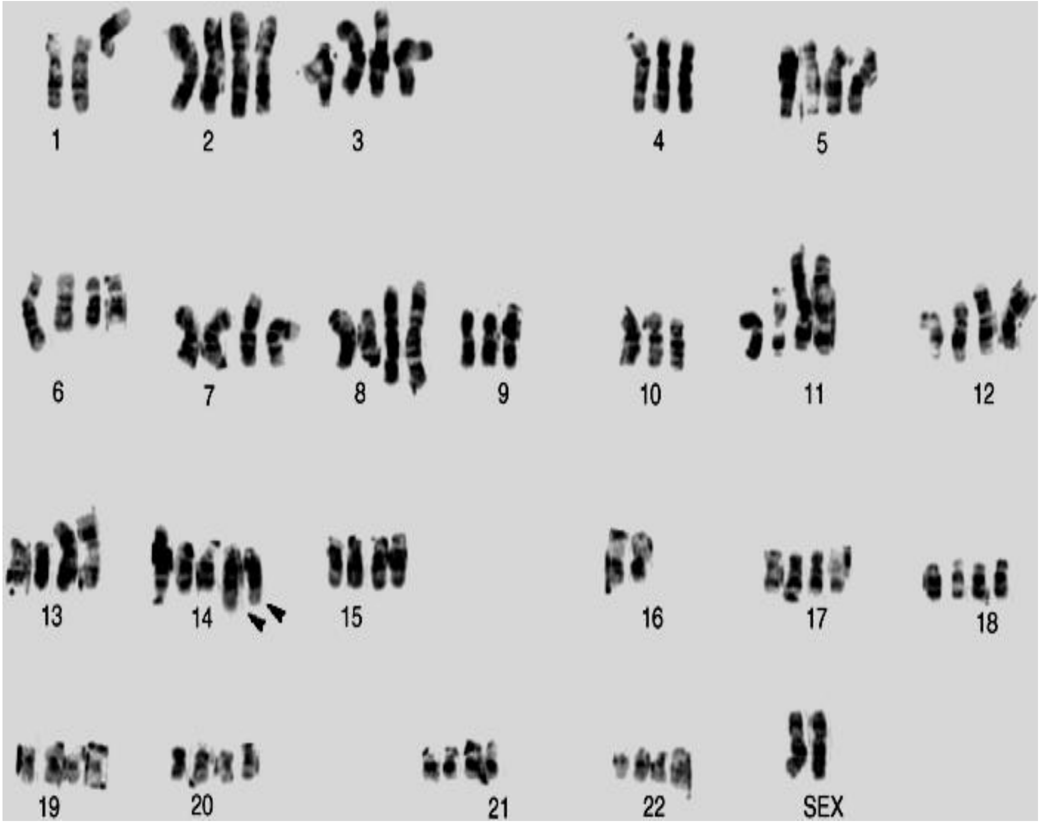
Triploidy



69, XXY

Triploidy observed in an abortus

Near Tetraploidy in Cancer



Numerical Chromosome Abnormalities: Aneuploidy

- **Aneuploidy** (Chromosome number is not a multiple of 23)
  - Monosomy: Loss of a chromosome. Not compatible with life (Except monosomy X/Turner syndrome)
  - Trisomy: Presence of an additional chromosome [frequently confused with triploidy (69 chromosomes)]

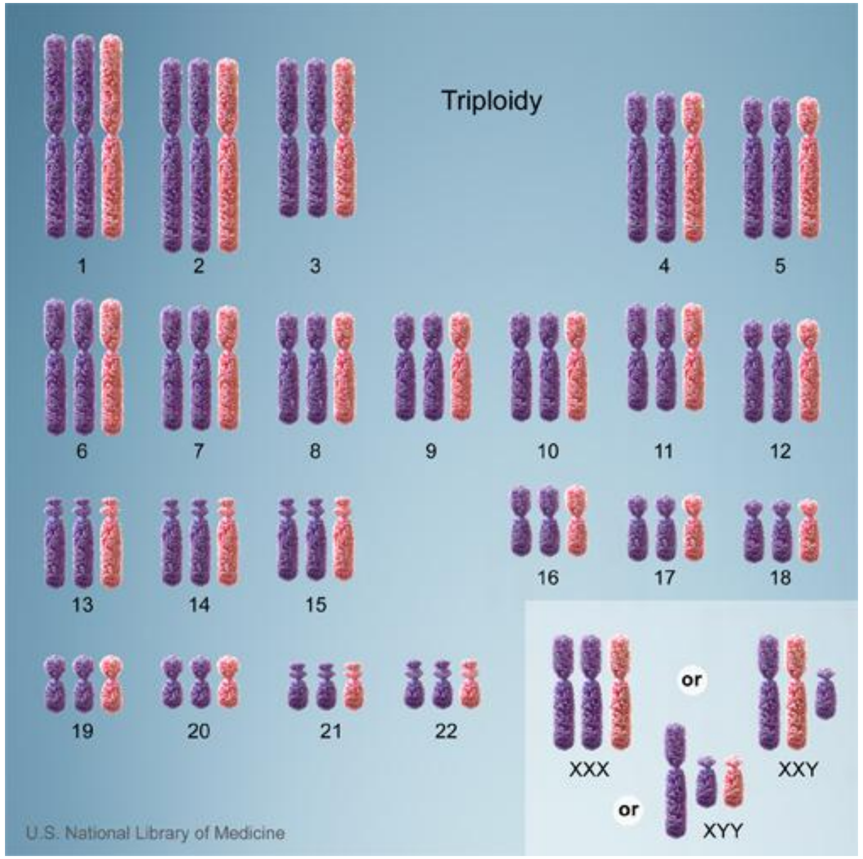
- Trisomy 21 (Down syndrome)
- Trisomy 18 (Edward syndrome)
- Trisomy 13 (Patau syndrome)

Autosomal trisomy
(Chromosome number 47)

- Turner Syndrome (45,X)
- Klinefelter Syndrome (47,XXY)

Sex Chromosome Aneuploidy

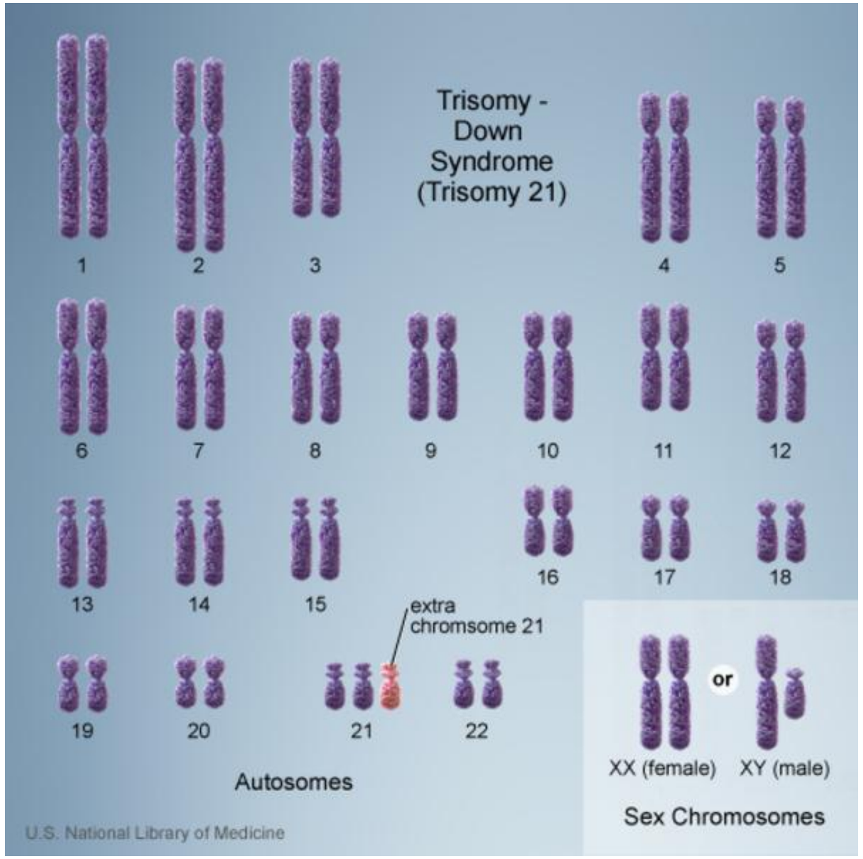
Triploidy



69,XXX or 69XXY or 69XYY

Not compatible with life; Spontaneous Abortions

Trisomy



47,XX or 47XY

Trisomy 21 (Down syndrome)

- Most common liveborn autosomal trisomy
- Risk factor: Increased maternal age, increases risk of meiotic nondisjunction during oogenesis (most common is meiosis I nondisjunction)
- Features:
  - Intellectual disability
  - Short stature
  - Depressed nasal bridge, upslanting palpebral fissures, epicanthal folds
  - Congenital heart defects
  - Single palmar crease
  - Develop changes similar to Alzheimer disease at a relatively young age.
    - One of the genetic factors responsible for Alzheimer is localized to chromosome 21 (amyloid precursor protein *APP*)

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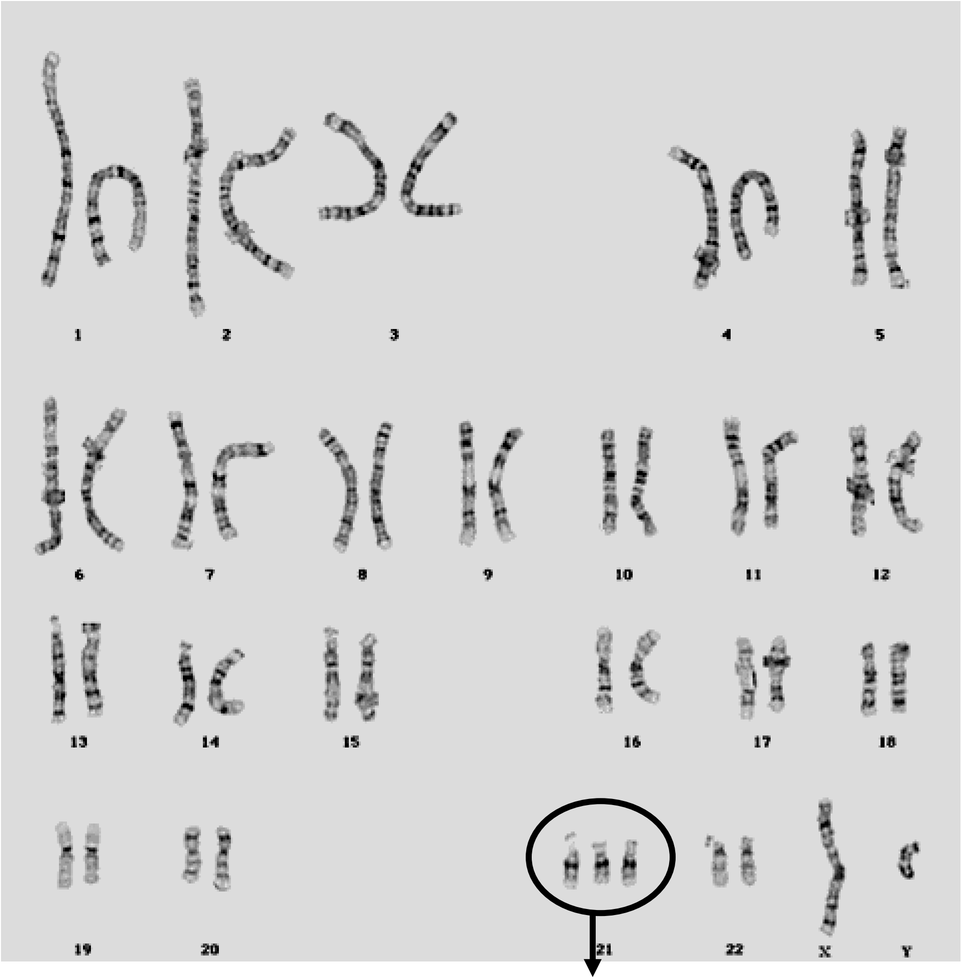
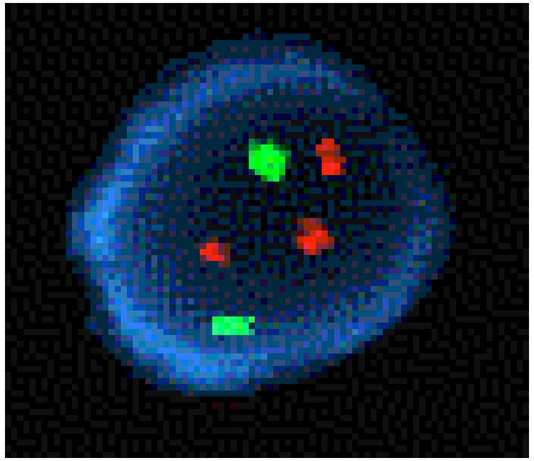
Karyotype of Trisomy 21



47,XY,+21

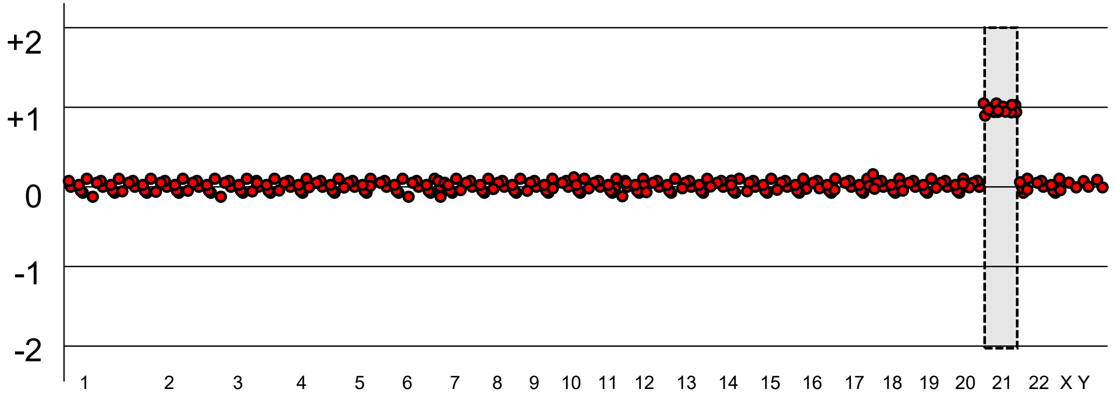
Chromosome 21



FISH probes will work with interphase stage cells to allow rapid return of results

- 2 chr 13 signals (green)
- 3 chr 21 signals (red)

- Prenatal diagnosis with maternal serum screening and ultrasound
- **Karyotype analysis or FISH with fetal cells is confirmatory**



Array CGH of all autosomes showing trisomy 21

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Down syndrome – genetic causes

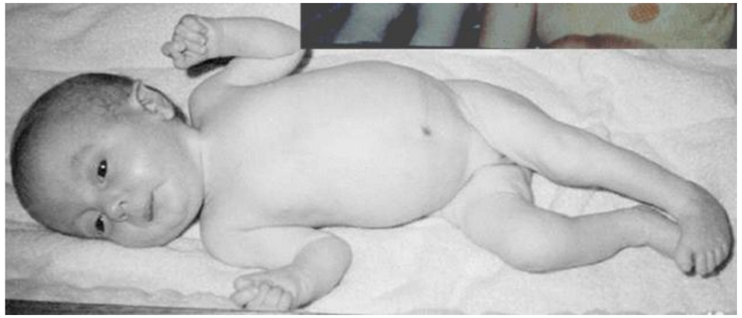
1. **Trisomy 21 - Due to non-disjunction during meiosis**
2. Some children may be mosaic – Some cells have trisomy 21; some other cells may have normal karyotype
3. *Down syndrome may also result from Robertsonian translocation*

Trisomy 18 (Edward syndrome)

Genetic mechanism is nondisjunction during **oogenesis**

Features:

- Clenched fist, overlapping of fingers
- Rocker bottom feet
- Congenital heart defects
- Low-set ears, small lower jaw (micrognathia)
- Microcephaly
- Intellectual disability



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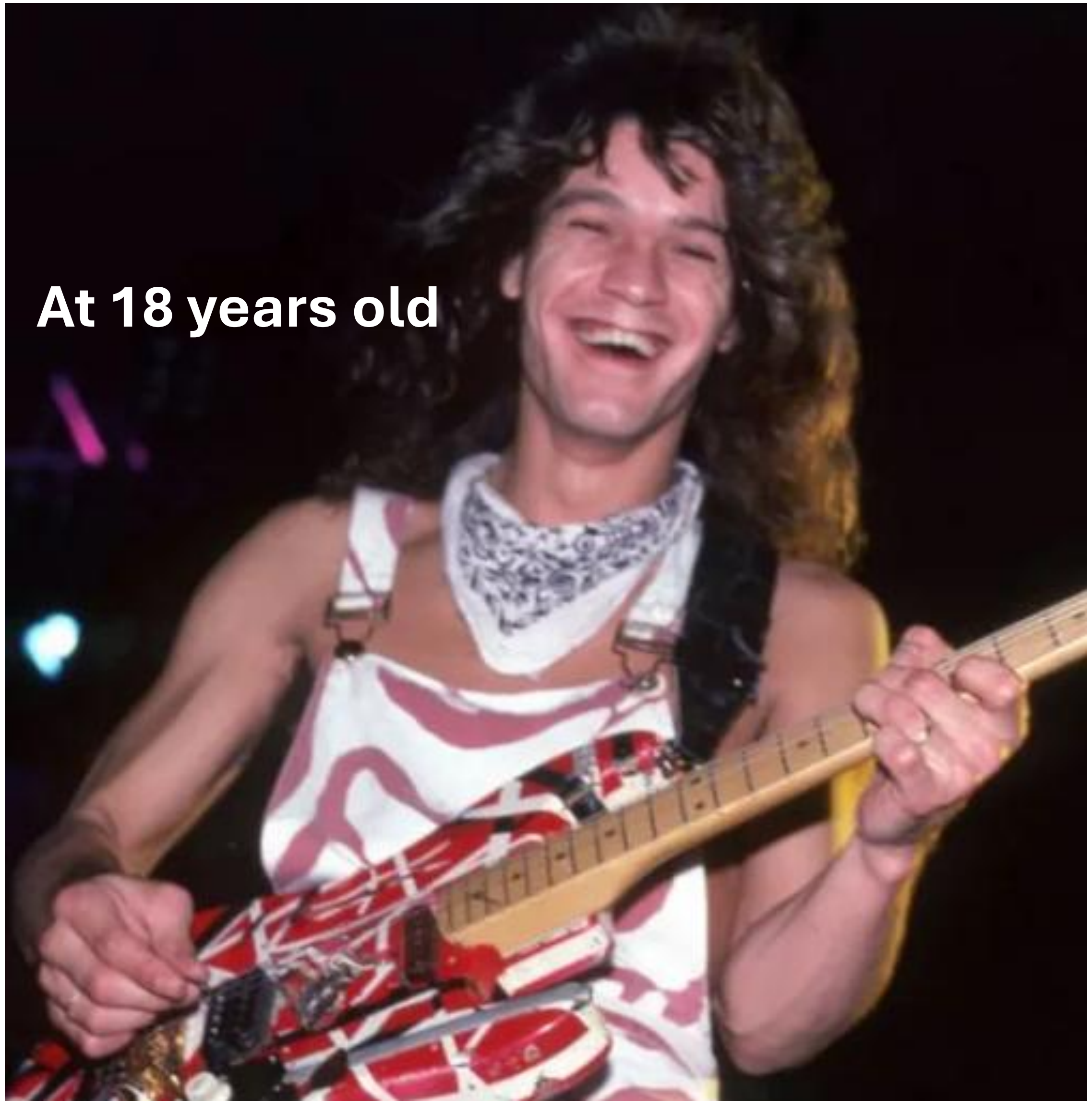
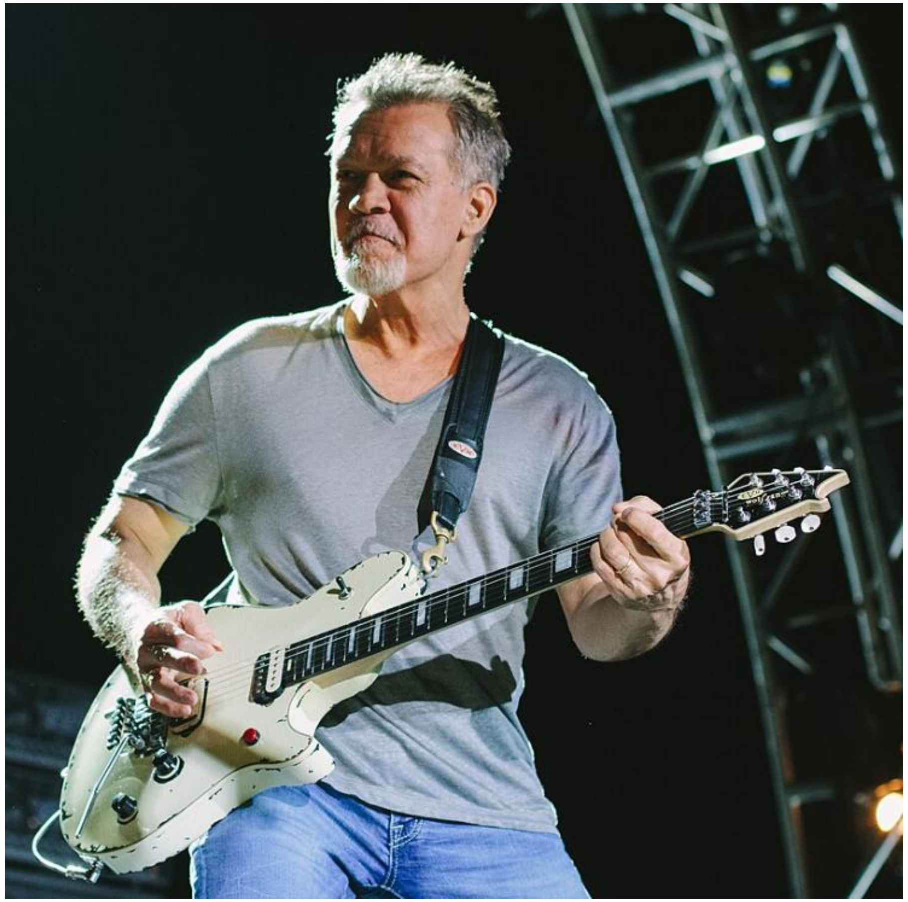


Eddie's **overlapping fingers**
Rolling stone: 4^{th} Greatest guitarist of all time
Rock and roll: **Rocker-bottom feet**



Eddie Van Halen

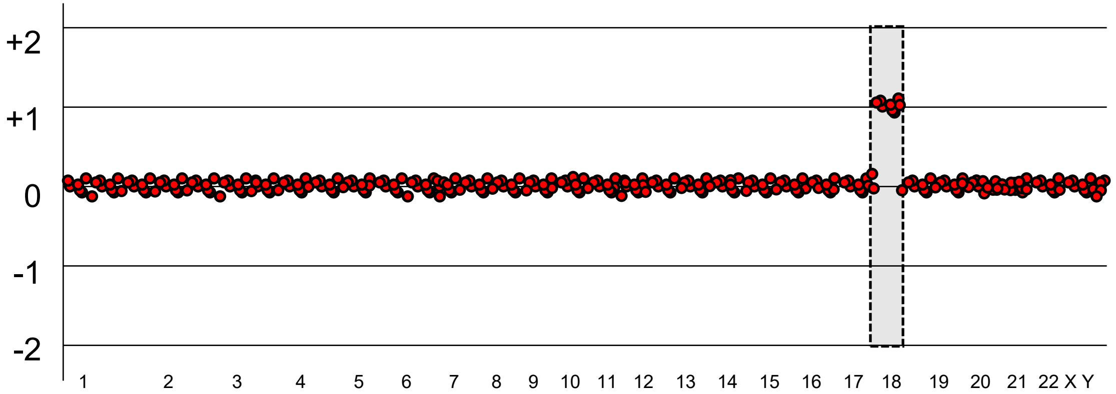
Karyotype of Trisomy 18

47,XY,+18

Chromosome 18

FISH probes will work with interphase stage cells to allow rapid return of results

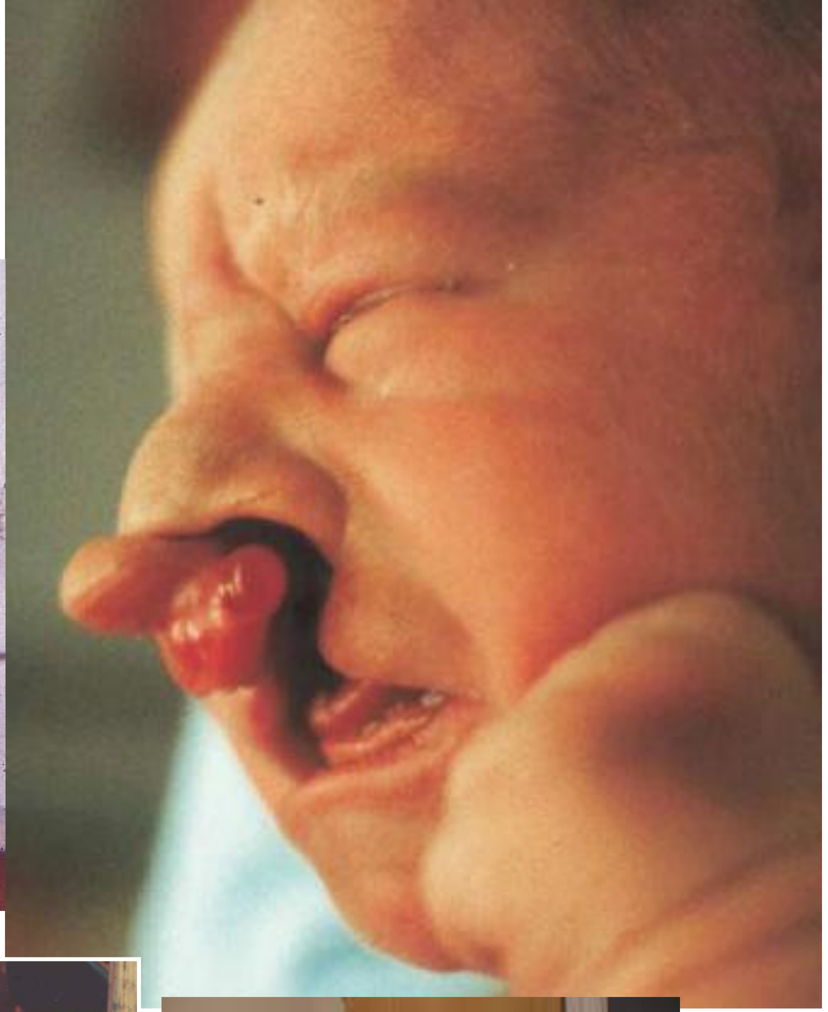
- 3 chr 18 signals



Array CGH of all autosomes showing trisomy 18

Trisomy 13 (Patau syndrome)

Most common genetic mechanism is nondisjunction during oogenesis



Features:

- Polydactyly
- Cleft lip and palate
- Microphthalmia
- Microcephaly
- Intellectual disability
- Cardiac anomalies (VSD or ASD)

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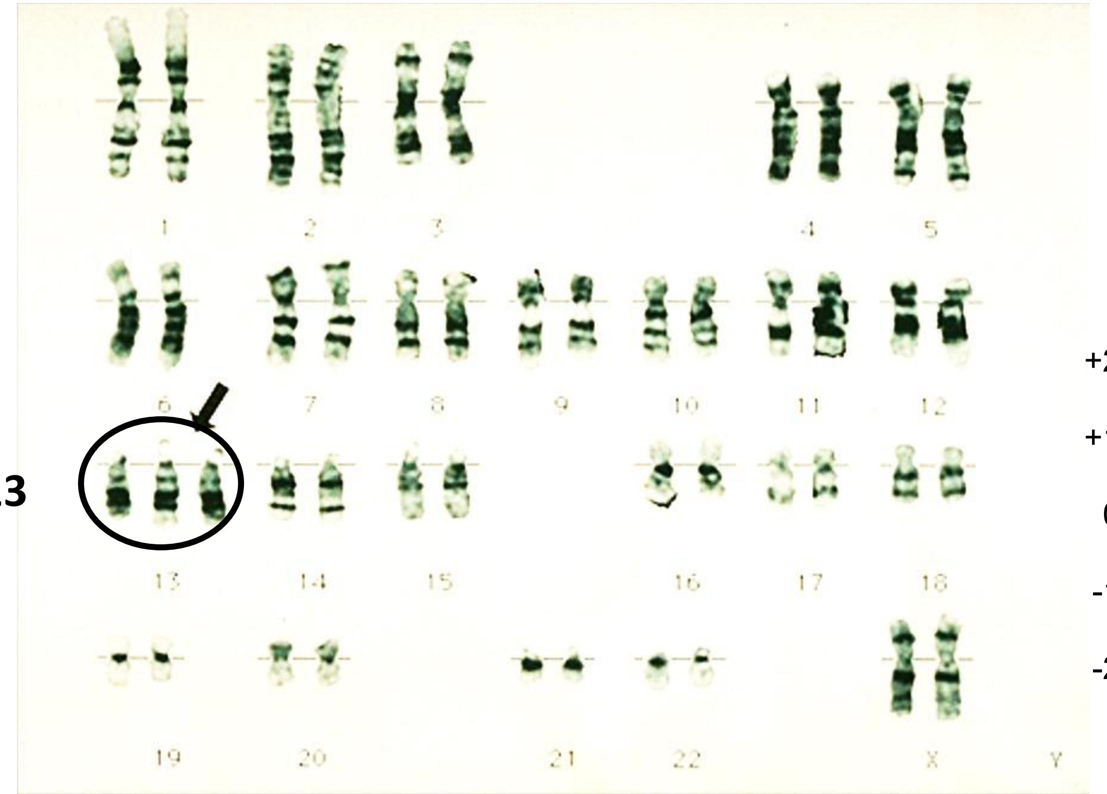
Karyotype of Trisomy 13

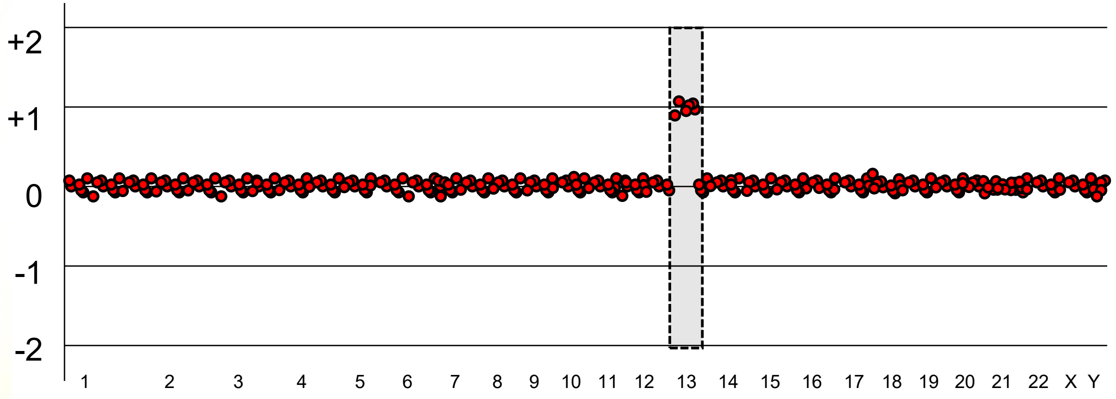
Chromosome 13



47,XX,+13

FISH probes will work with interphase stage cells to allow rapid return of results

- 3 chr 13 signals



Array CGH of all autosomes showing trisomy 13

Sex Chromosome Aneuploidy

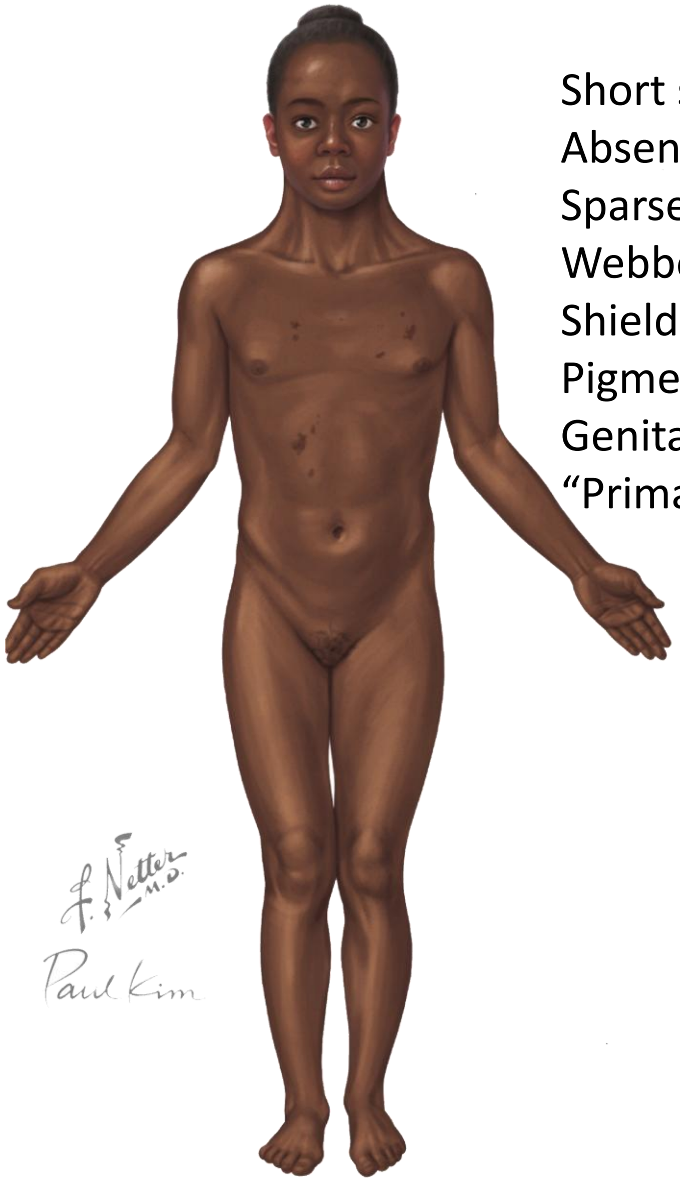
- Turner Syndrome (45,X)
- Klinefelter Syndrome (47,XXY)

Turner Syndrome (45,X)

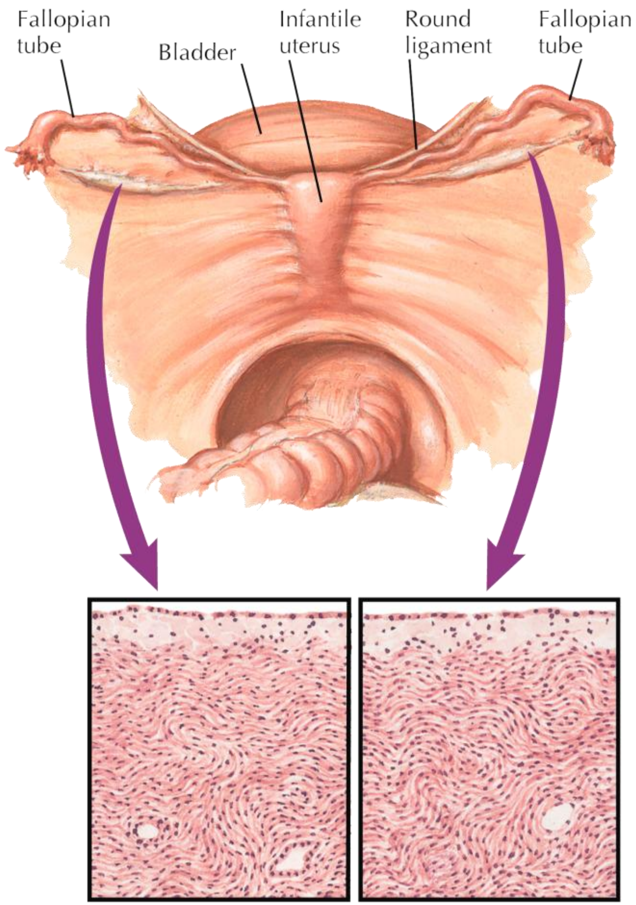
- X chromosome monosomy
  - Usually due to nondisjunction during ***spermatogenesis***
- Short stature
- Webbed neck, cystic hygroma at birth (neck swelling)
- Primary amenorrhea
- Gonadal dysgenesis
- ‘Streak ovaries’
- Lymphoedema of hands and feet

://www.dailyhome.com/view/full_story/5548416/article-A-unique-teenager

Turner Syndrome (45,X)



Short stature
Absence of secondary sex characteristics
Sparse pubic hair
Webbed neck
Shield-like chest
Pigmented nevi
Genital streaks in place of gonads
“Primary amenorrhea”



Primitive genital streaks in place of gonads, wavy stroma with absence of germinal elements

Therapies:

- Hormone replacement therapy
- Human growth hormone if diagnosis established before 10-years-old
- Assisted reproduction with donor egg is advancing

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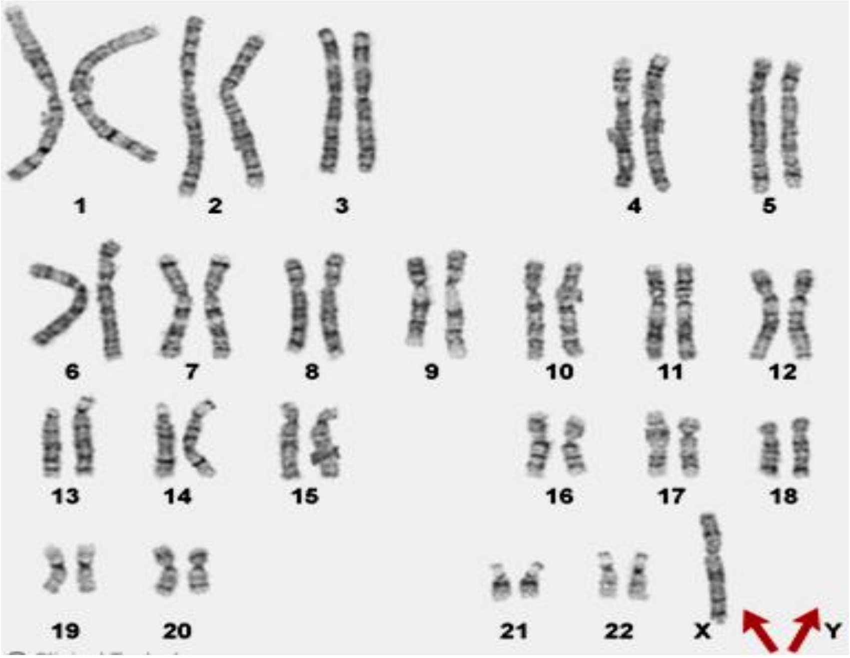
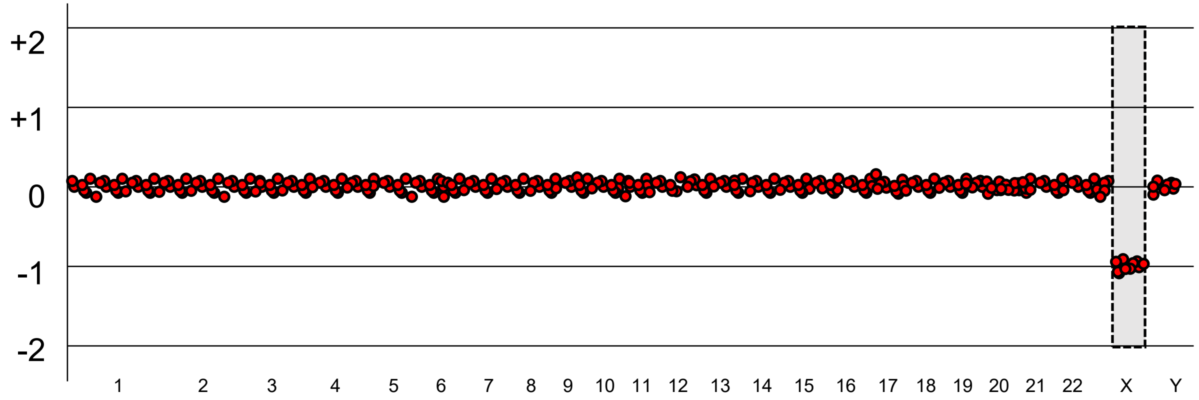
Turner syndrome karyotype (45,X)

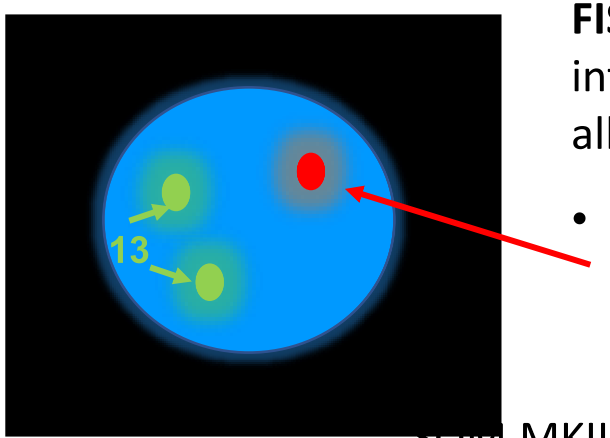


45,X

- Many TS patients are mosaic
  - Some of their cells are 45,X
  - Other cells are 46,XX and 47,XXX (indicate mitotic nondisjunction during embryogenesis)
- Cells with 45,X: Have **no Barr body** (Inactivated X-chromosome) **(Female with NO Barr body)**
- Isochromosome X will also lead to Turner syndrome (*discussed next lecture*)



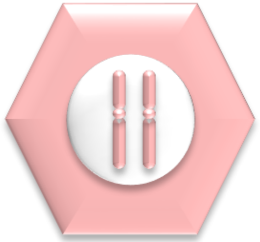
Array CGH of all chromosomes showing monosomy X (control DNA is female)



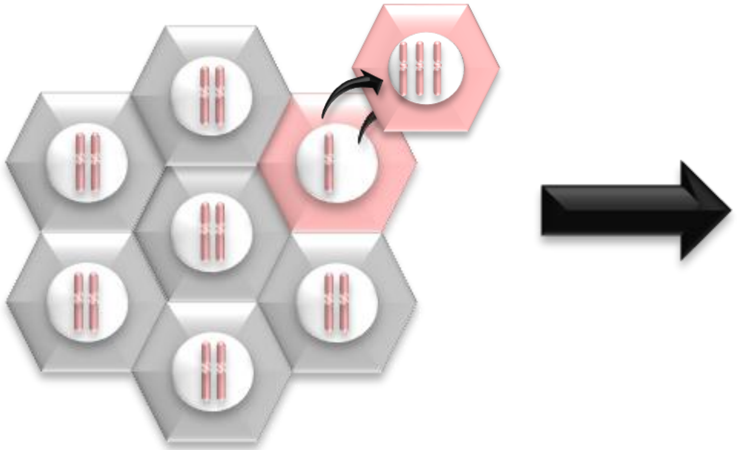
FISH probes will work with interphase stage cells to allow rapid return of results

- 1 chr X signal

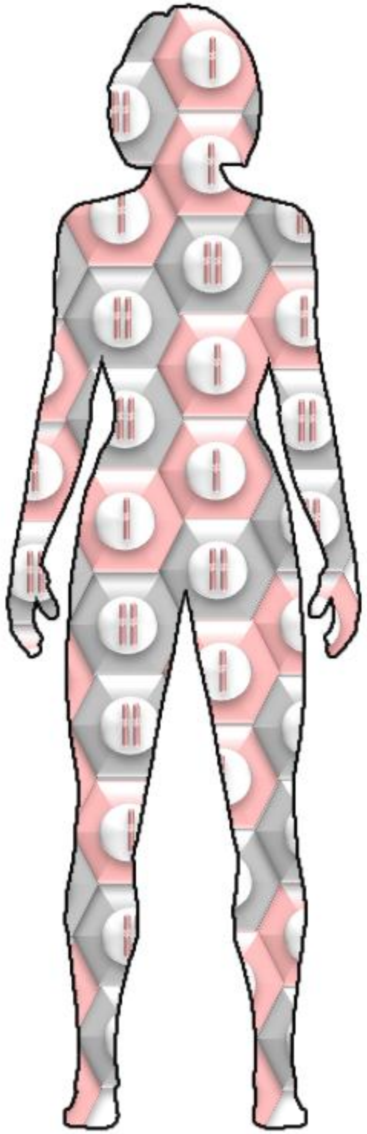
Mosaic Turner Syndrome- 46, XX



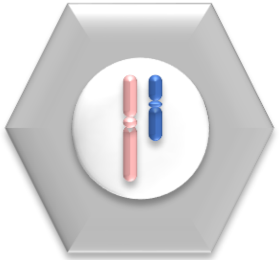
- **Some females are mosaic for Turner Syndrome**
  - Fertilized egg, 46,XX
  - Mitotic Nondisjunction during embryogenesis
  - Some cells are 45,X others are 46,XX and 47,XXX



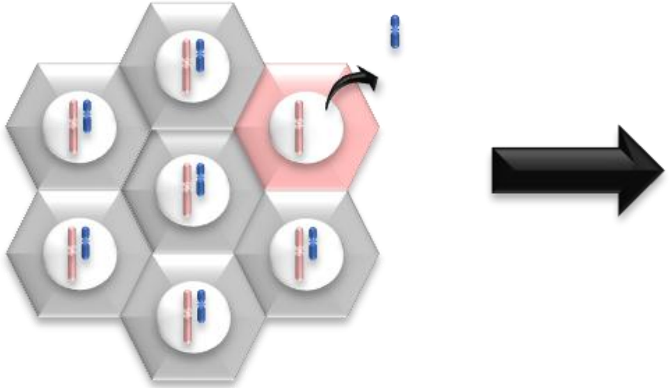
During a post-fertilization mitosis, one of the X chromosomes evicted during cell division



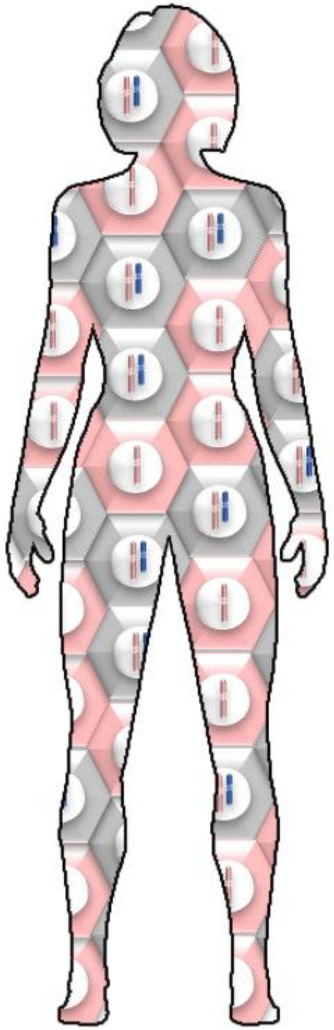
Mosaic Turner Syndrome- 46,XY



- **Some phenotypic females are mosaic**
  - Fertilized egg 46,XY
  - Mitotic Nondisjunction during embryogenesis
  - Some cells are 45,X others are 46,XY
  - **Mosaic 45,X/46XY are recommended for**
  - **oophorectomy due to risk of gonadoblastoma**



During a post-fertilization mitosis, the Y chromosome is evicted during cell division

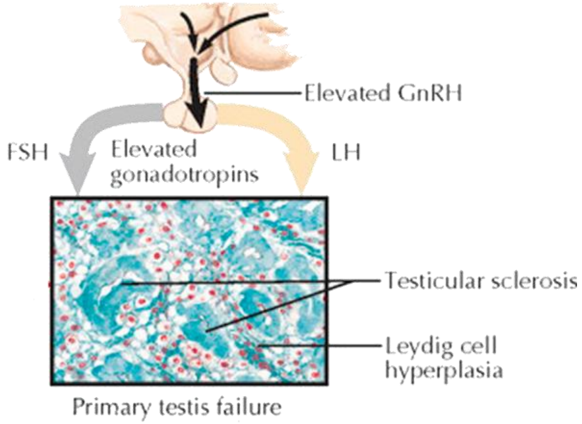


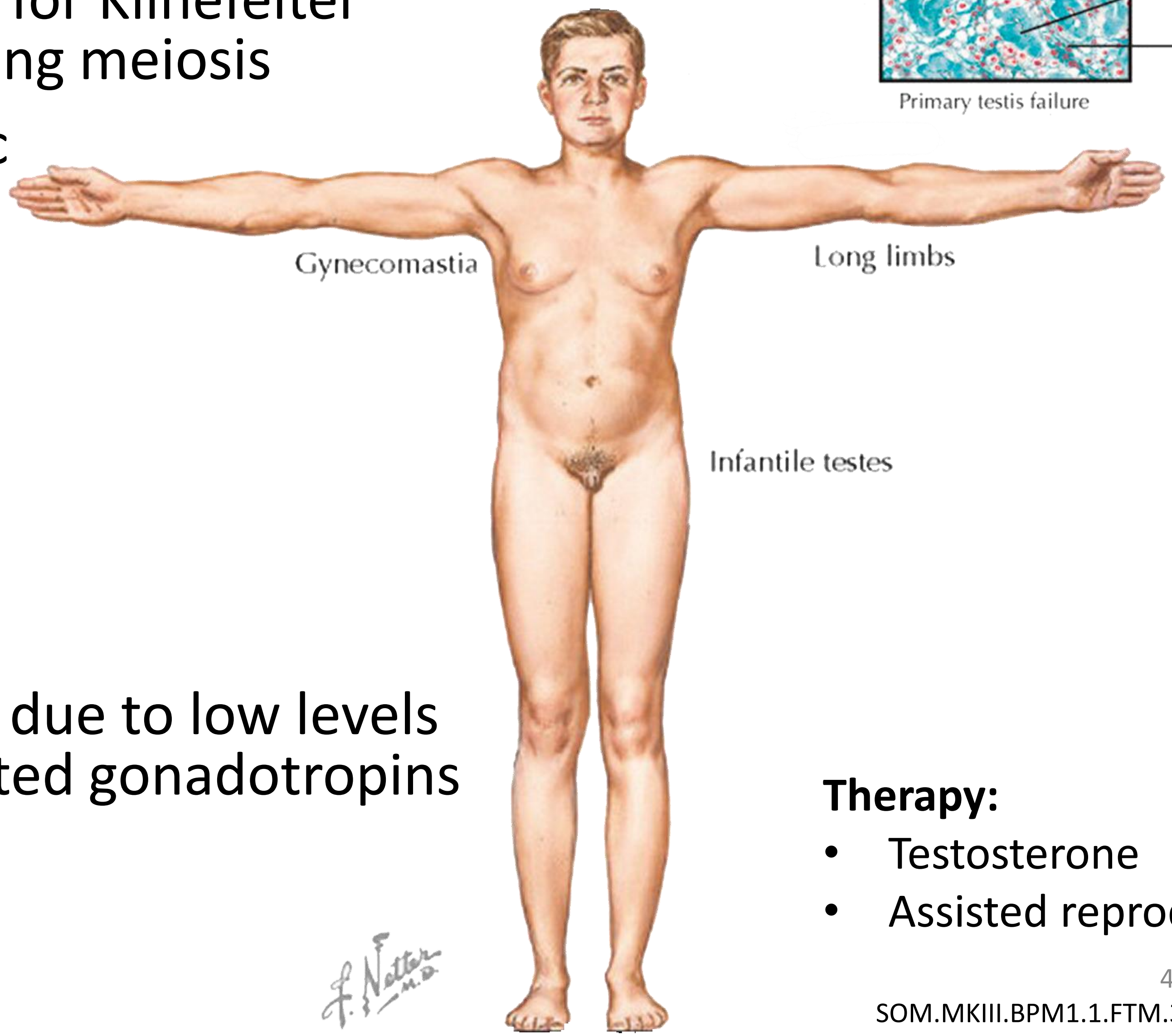
Klinefelter syndrome (47,XXY)

- Genetic mechanism responsible for Klinefelter syndrome is nondisjunction during meiosis
- Some individuals may be mosaic
    - (46XY/47XXY/ 48XXXY)

Features:

- Tall with long limbs
- Gynecomastia
- Female distribution of hair
- Testicular atrophy and infertility due to low levels of testosterone leading to elevated gonadotropins
- Feminization of fat distribution
- Osteoporosis

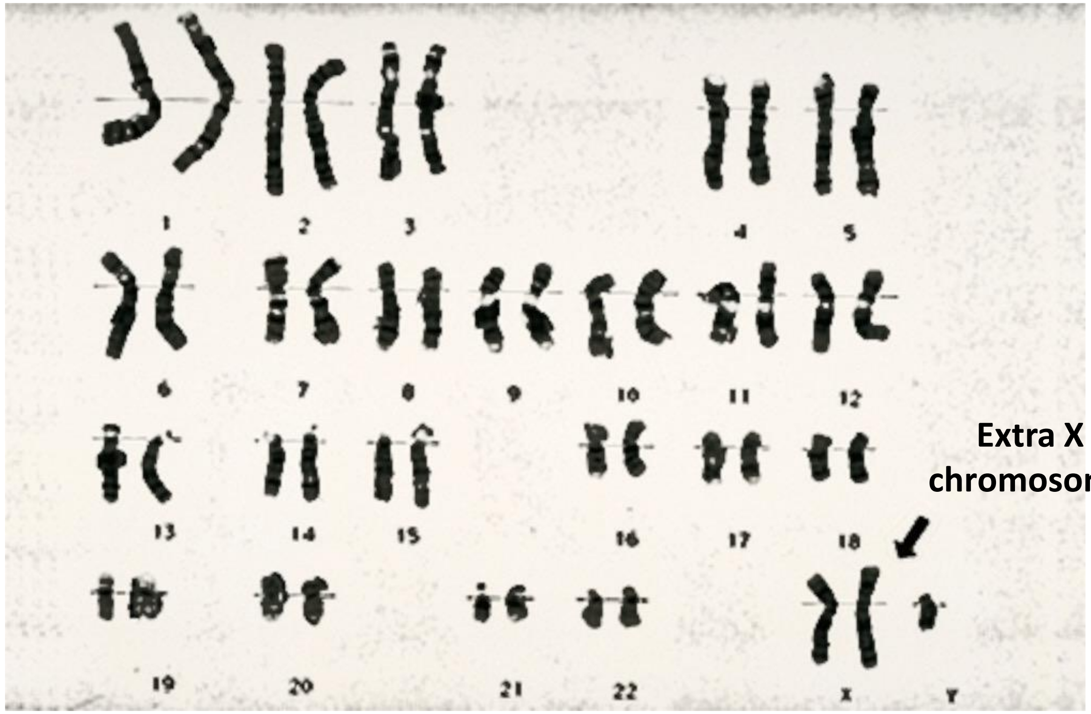




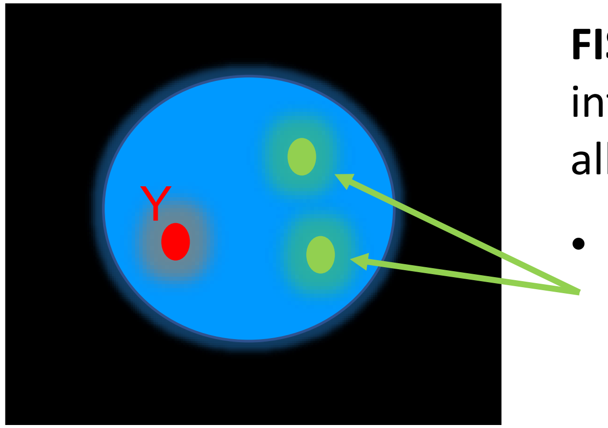
Therapy:

- Testosterone
- Assisted reproduction

Klinefelter Syndrome (47,XXY)

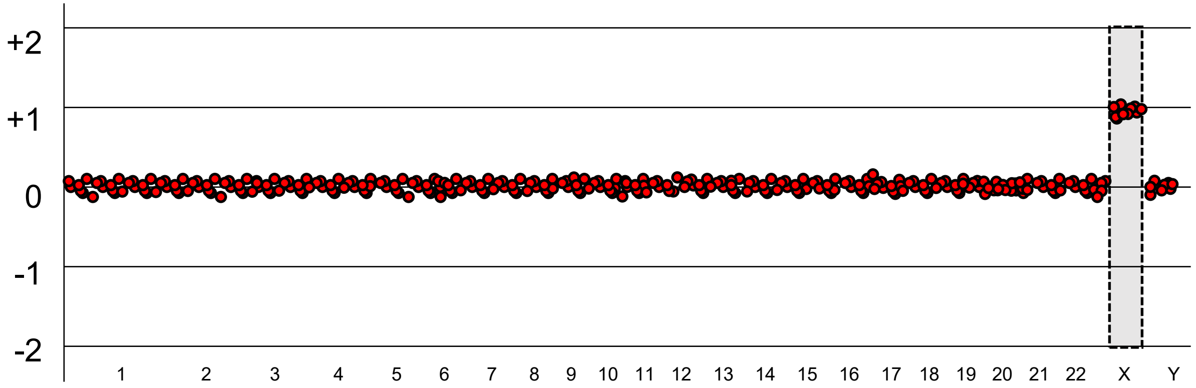


47,XXY



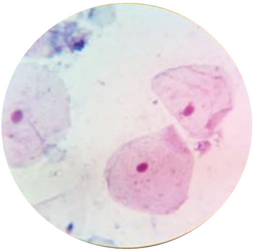
FISH probes will work with interphase stage cells to allow rapid return of results

- 2 chr X signals

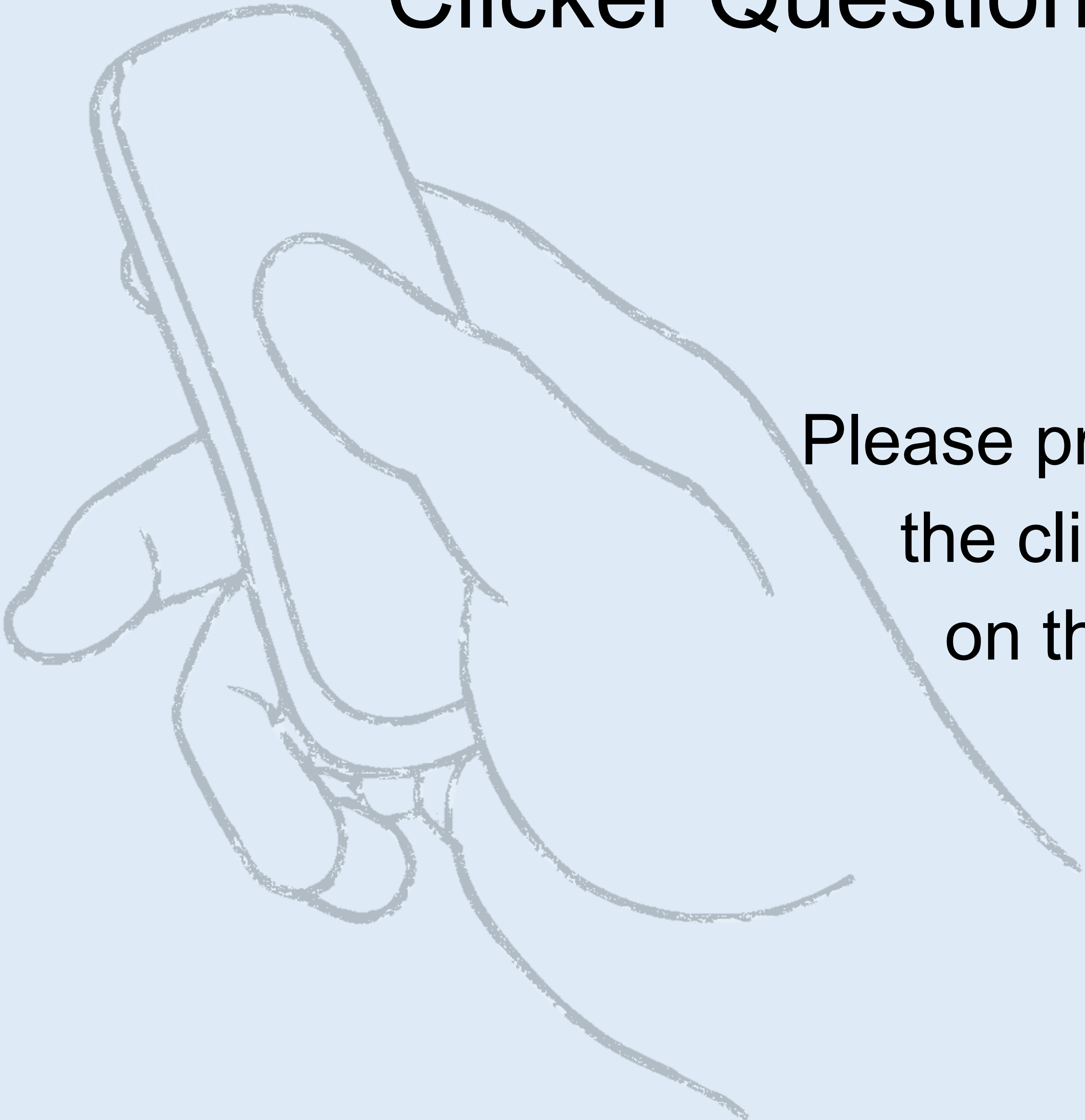


Array CGH of all chromosomes showing extra X (control DNA is male)

Presence of a Barr body (X- chromosome inactivation) will be observed in the buccal mucosal cells



Clicker Question Next!



Please prepare to answer
the clicker question
on the next slide

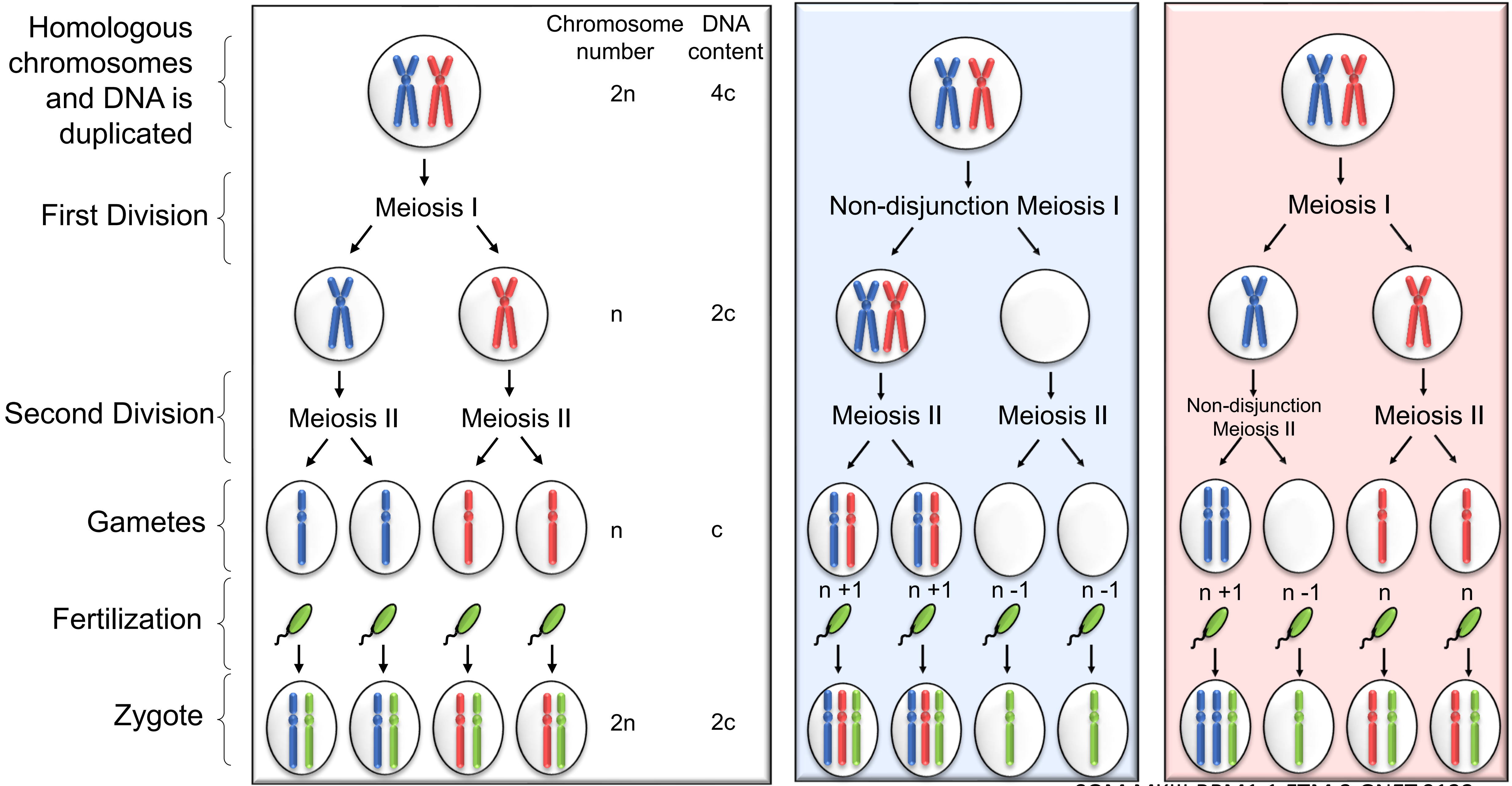
Chromosomal Aneuploidy is thought to occur by Nondisjunction

• Trisomy 21	1/830 live births
• 47,XXY (Klinefelter)	1/1000 live births
• 45,X (Turner)	1/4000 live births
• Trisomy 18	1/7,500 live births
• Trisomy 13	1/22,700 live births

You do not need to memorize these numbers!

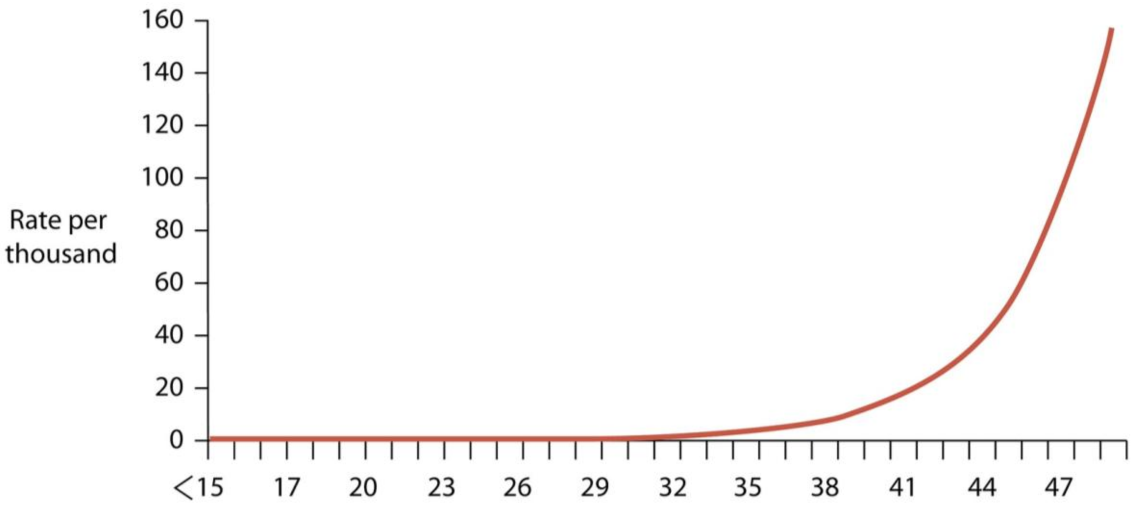
Data are approximated from Hsu LYF: Prenatal diagnosis of chromosomal abnormalities through amniocentesis, 1998

Meiosis and non-disjunction



Non-disjunction

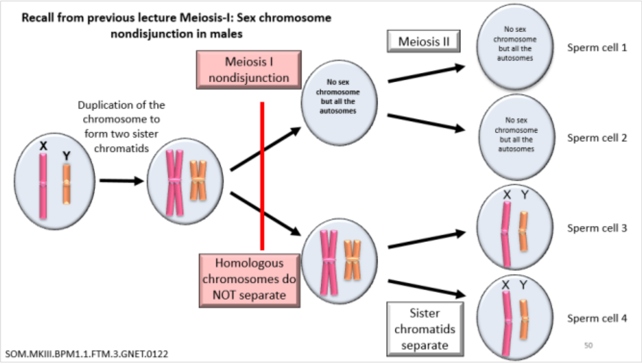
- Non-disjunction is most common in female **meiosis I**
- Individuals with Down syndrome usually have two maternal chromosomes 21 and one paternal chromosome 21.
- If two maternal chromosomes are not identical to each other, the non-disjunction occurred at meiosis I (that is they have different alleles shown as blue and red on previous slide)
- If the two maternal chromosomes are identical, the non-disjunction occurred at meiosis II (sister chromatid shown as two blue alleles on previous slide)

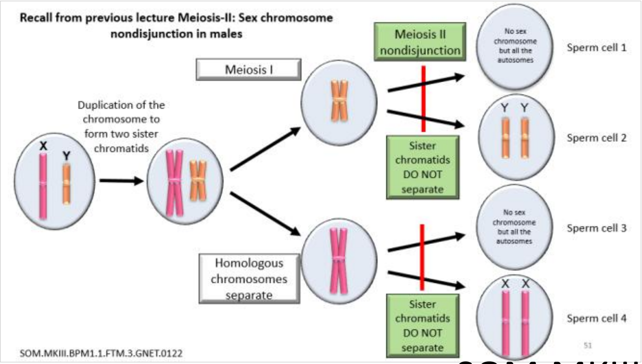


Increase in the risk of nondisjunction resulting in trisomy 21 with advanced maternal age

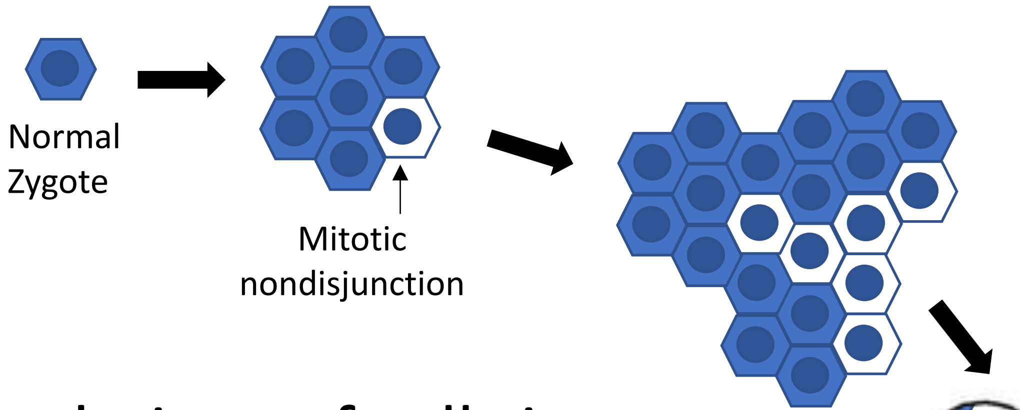
Draw out which type of non-disjunction, *meiosis I or meiosis II*?

- 47, XXY Klinefelter syndrome may be caused by nondisjunction during meiosis I or II in the mom, or nondisjunction during meiosis I in dad
  - Explain why meiosis II nondisjunction in the father (sperm) cannot result in Klinefelter syndrome (hint: a person with Klinefelter syndrome is male)
- 47, XYY fetus is the result of meiosis II nondisjunction in dad
  - May not be caused by meiotic non-disjunction in the mom (hint: this person is also male)
- 47, XXX females may be the result of meiosis I or II nondisjunction in moms or meiosis II non-disjunction in dad
  - May not be caused by meiosis I non-disjunction in the dad





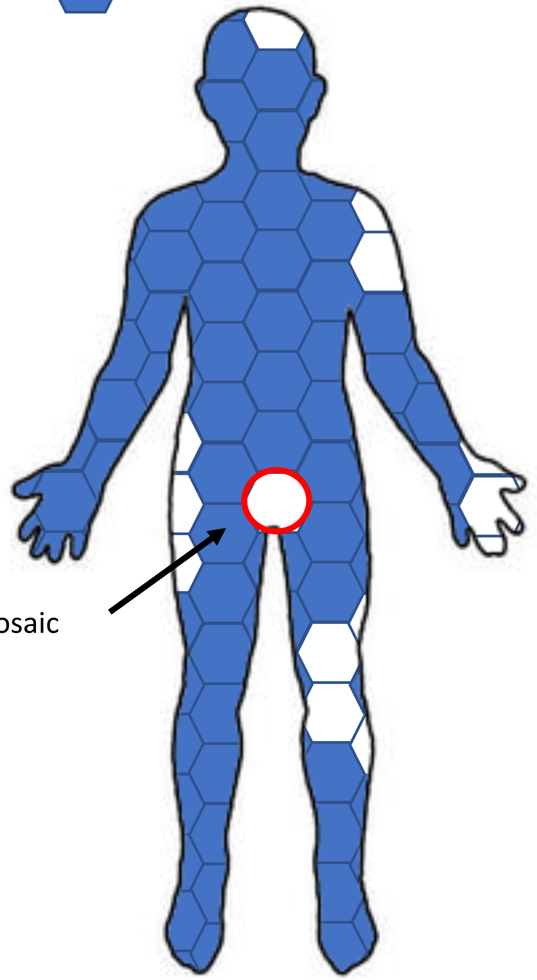
MOSAICISM



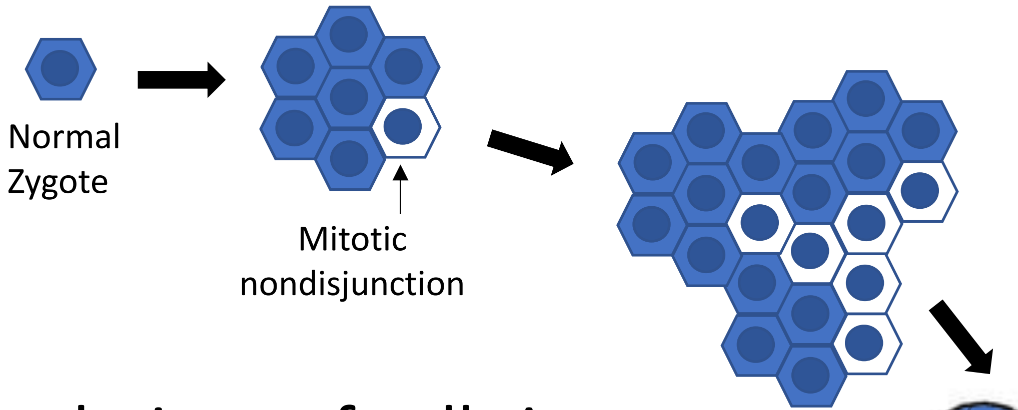
- The presence of two or more populations of cells in one individual with different genotypes

Mixed somatic/germline mosaic

Some percentage of the cells in a person body are of one genotype, and other cells are of another genotype



MOSAICISM



- The presence of two or more populations of cells in one individual with different genotypes

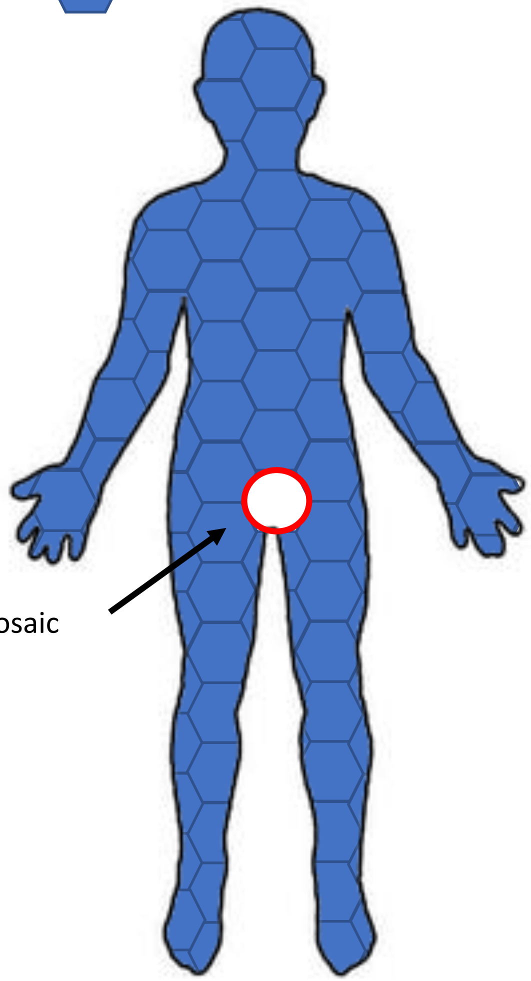
Mixed somatic/germline mosaic

Some percentage of the cells in a person body are of one genotype, and other cells are of another genotype

Confined germline mosaic

All cells in the body are of one genotype

Cells of the gametes are something else (leads to high risk of having children with a genetic disorder)

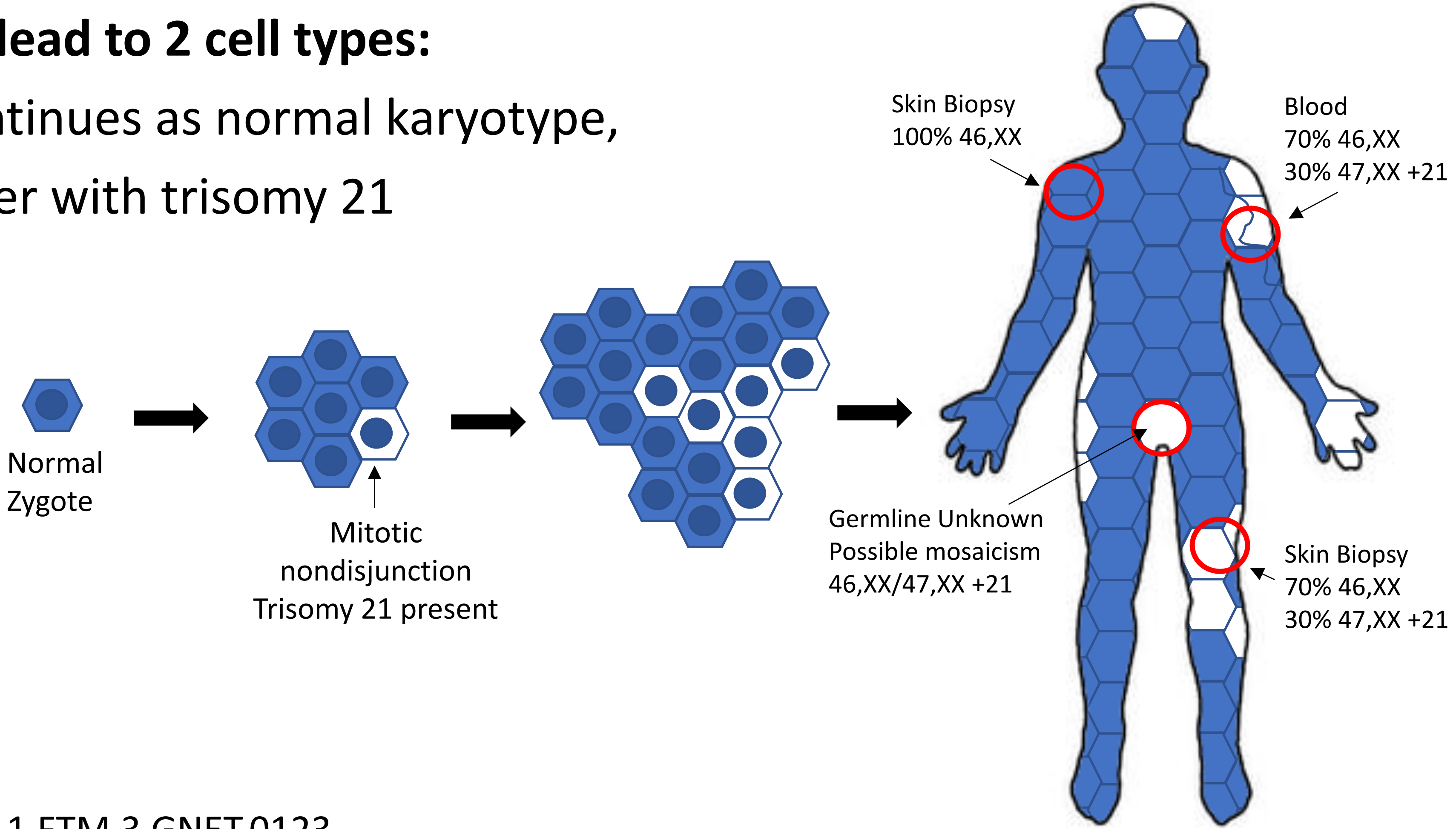


Consider 2 ways that a person could be mosaic for a Down syndrome:

1. Non-disjunction may occur as a **post-zygotic** event during embryonic development; that is mitotically (after fertilization of a normal ovum and normal sperm)

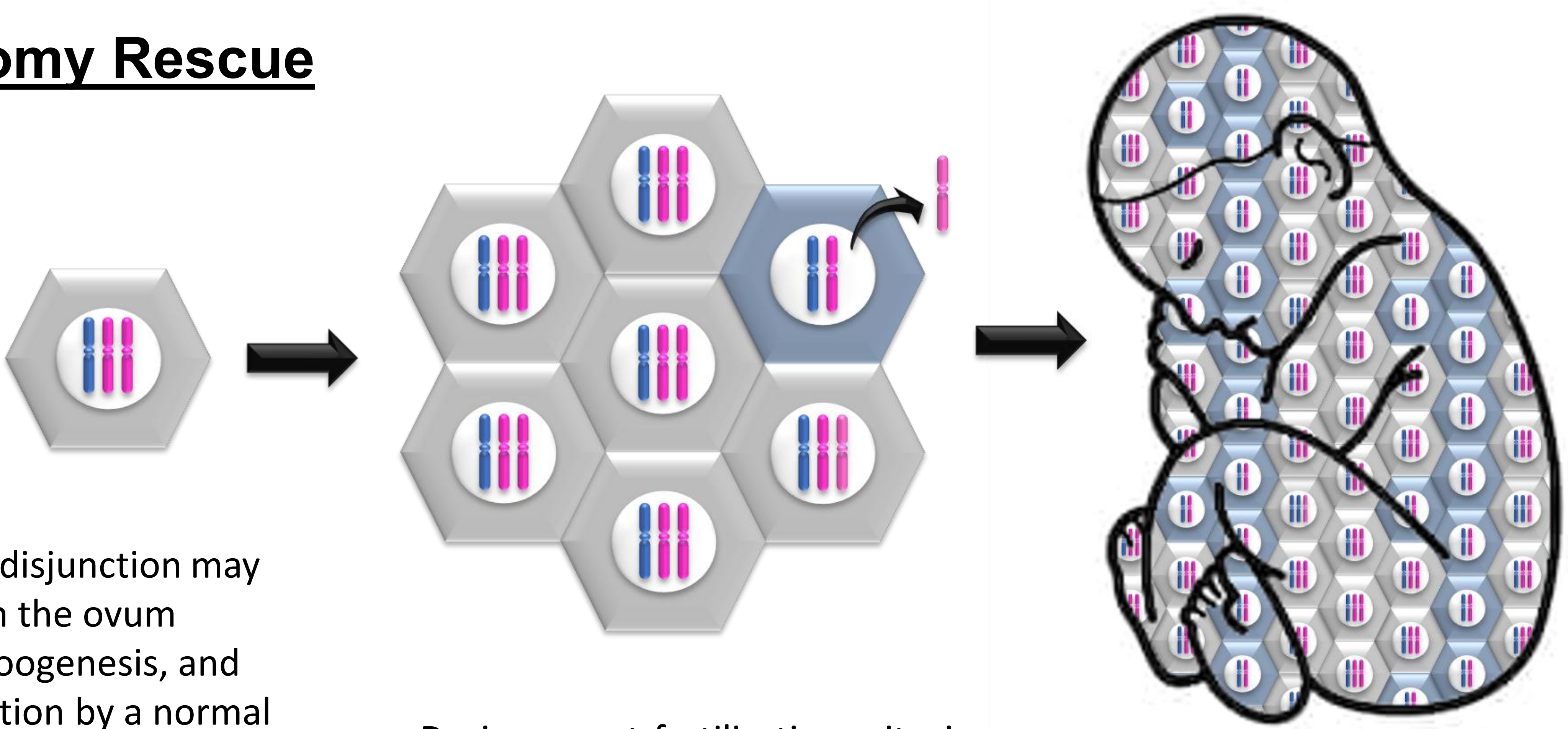
This will lead to 2 cell types:

- one continues as normal karyotype,
- the other with trisomy 21



Consider 2 ways that a person could be mosaic for a Down syndrome:

Trisomy Rescue



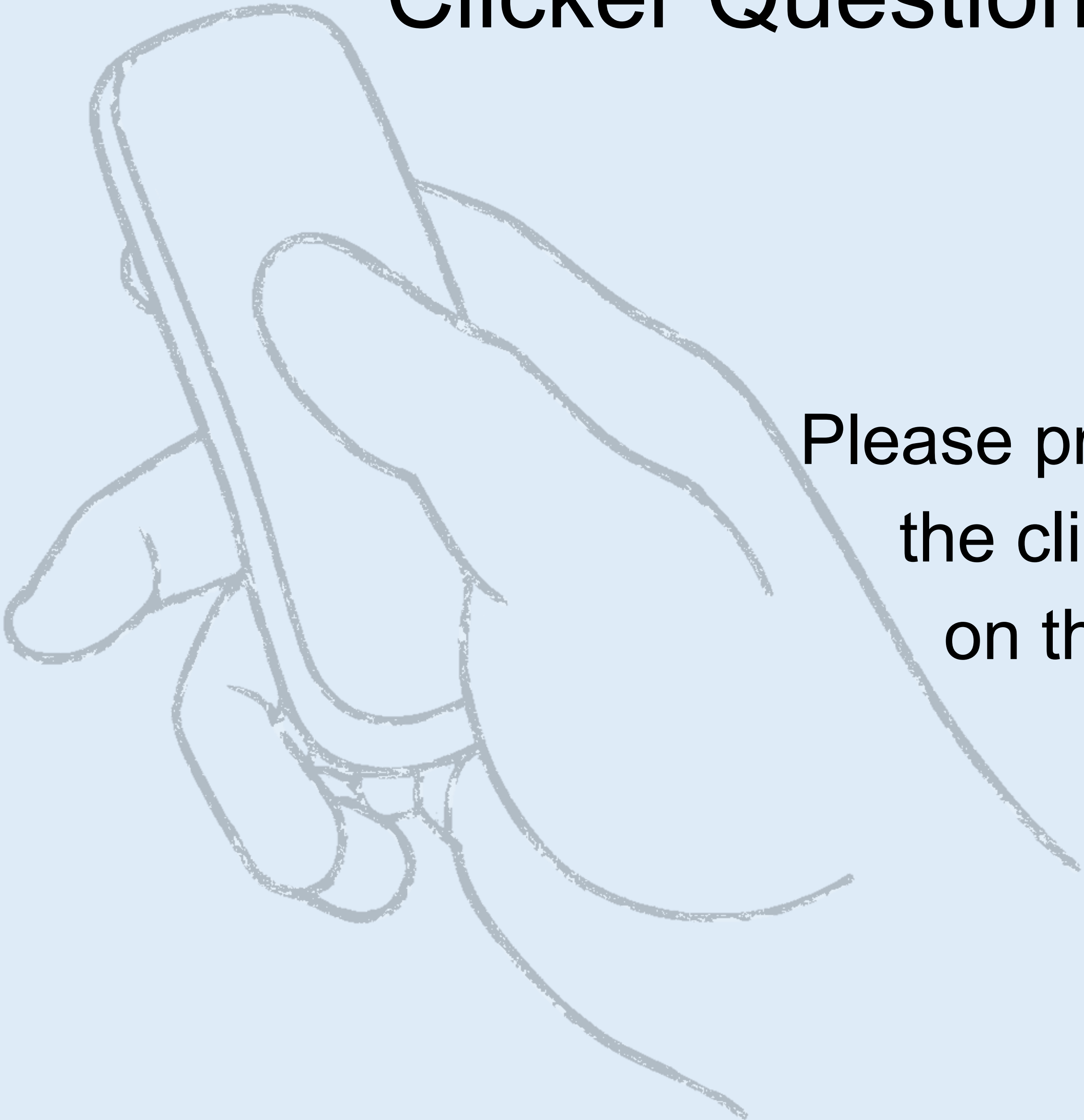
2. Non-disjunction may occur in the ovum during oogenesis, and fertilization by a normal sperm leads to trisomy 21 in the conception

During a post-fertilization mitosis, one of the extra chromosome 21 is evicted during cell division

Now the fetus has 2 cell types, one with normal karyotype, and one that is trisomy 21

Don't run away, clicker coming next

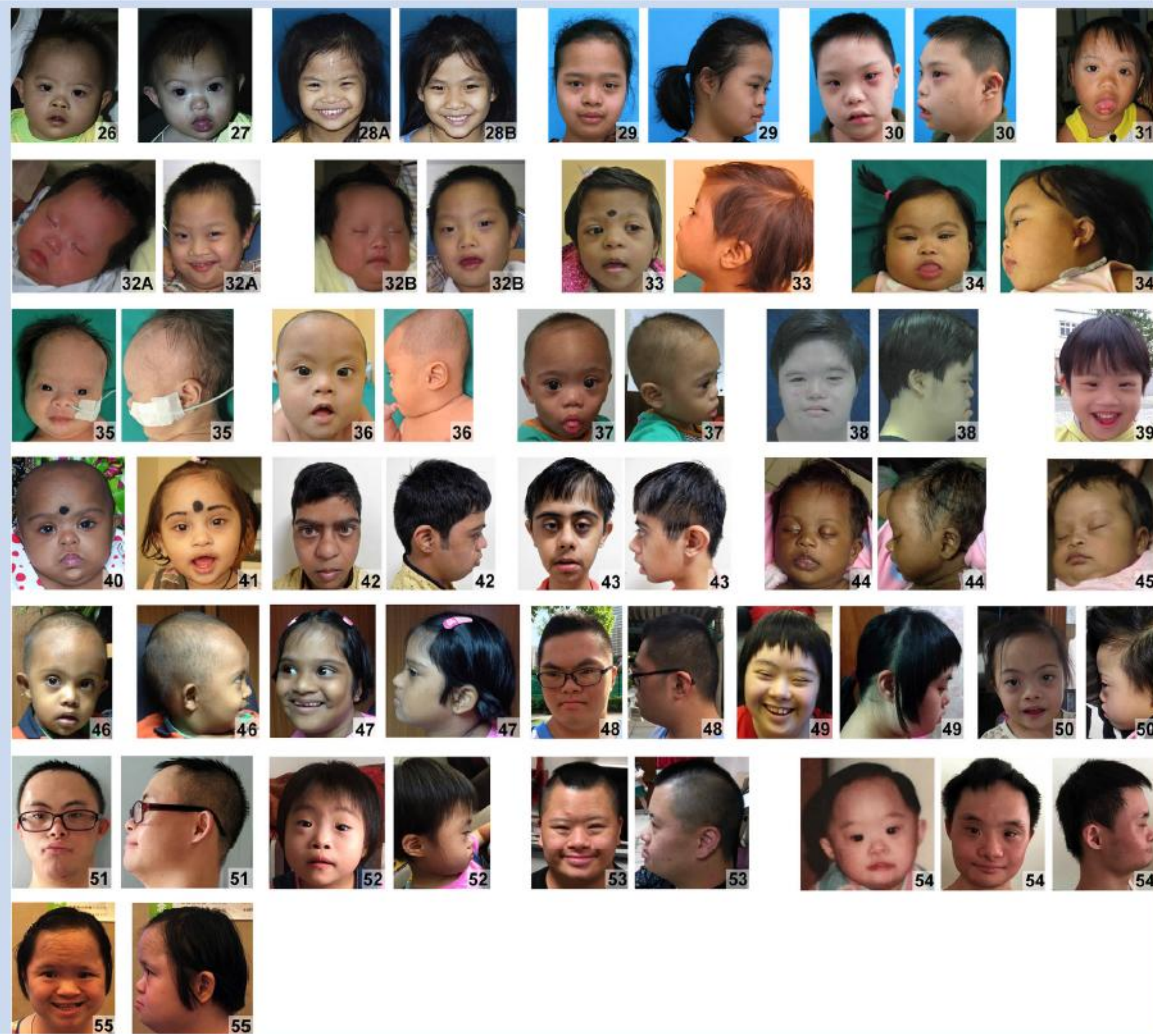
Clicker Question Next!



Please prepare to answer
the clicker question
on the next slide

Down Syndrome in Diverse Populations

Have a great day



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